

GENERAL SPECIFICATION G2.23

Health, Safety and Environment

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G2.23 HEALTH, SAFETY AND ENVIRONMENT

G2.23.1 Purpose

- (a) This specification defines the overall Requirements of the Contractor's health, safety and environmental (HSE) systems and procedures. The Contractor shall refer to General Specification G9 for specific requirements for environmental.

G2.23.2 Objectives, Strategies and Safety Mission Statement

(a) Objectives

- (i) The HSE objectives of the DTSS2, which are applicable to the Contract, can be summarised as follows:
- 1) To strive towards an Incident-Free environment with:
 - No loss of life;
 - No injury to persons;
 - No damage to property; and
 - No damage to the environment.
 - (ii) To promote a positive Safety Culture across the project through:
 - 1) Client and management commitment;
 - 2) Sharing safety moments, observations and lessons learned; and
 - 3) Recognition and rewards for exemplary HSE performance.
 - (iii) The Project is committed to promote an Incident-Free (IF) environment, a mind-set intolerable of incidents of any form, frequency or severity which harms personnel and/or the environment.

(b) Strategies

- (i) Achieving the safety objectives will require much effort by all stakeholders. With that in mind, the following key strategies are intended to reach out to all involved in the project at the various levels. Four key strategies developed towards this effort include:
- 1) Commitment by project leadership;
 - 2) Improving the competency levels of the workforce;
 - 3) Working collaboratively to deliver the Project safely; and
 - 4) Celebrating successes and giving due recognition to encourage working safely.

The Contractor project leadership shall take a leading role in defining the actions to be put in place to ensure these strategies are realised during construction, in close consultation with the Board and the S.O..

(c) Embrace the DTSS HSE Mission Statement

- (i) The DTSS 2 Safety Mission Statement states:
- 1) We aim for ZERO HARM and promote the well-being of everyone;

- 2) No injury is too small, and one injury is one too many;
 - 3) We will not be limited by past practices, and are committed to surpass current industry norms; and
 - 4) Every person on the DTSS 2 project has the responsibility and is empowered to take action on the issue of SAFETY.
- (ii) The Contractor shall embrace the DTSS 2 HSE Mission Statement.
 - (iii) The Contractor shall compose a mission statement for the Contract, in line with DTSS 2 Mission Statement.

G2.23.3 Legal Requirements

- (a) The Contractor shall allow in his rates and prices for complying with this Specification and Particular Specification, all applicable legislative HSE requirements of Singapore including any new acts, regulations and approved codes of practice which may be gazetted during the Contract Period including any amendments or re-enactments thereto not specifically mentioned in the Specifications.
- (b) The Contractor shall comply with the HSE Requirements contained in this Contract Specification even where these impose a higher standard than required by current Singapore legislation. In the event of conflict between the Specifications and the applicable laws and regulations whereby the Specifications would prevent compliance with the laws and regulations, the laws and regulations shall apply. In the event that the Specifications impose a higher or stricter standard than the laws and regulations, the Specifications shall apply.
- (c) The Contractor shall demonstrate that it is keeping up to date with changes in the approved code of practices by subscribing to Singapore Standards eShop or equivalent and other international standards referenced in the Contract.

G2.23.4 In-House Rules and Regulations

- (a) The Contractor shall post HSE notices and information prominently around its Site in appropriate languages for its workforce. Such notices shall comply with industry standards and legislation and shall be submitted to the S.O. for review and acceptance.
- (b) The Contractor shall comply with the Board's Life Saving Rules and shall ensure that these rules are fully briefed to the workforce and prominently displayed around the works. The S.O. will advise additional Project Life Saving Rules. The workforce shall be informed that infringement of any of these rules may result in immediate termination from the Project. This rule shall apply onto every Project member. The Board's Life Saving Rules are:
 - (i) Do Not Work without a valid Permit to Work;
 - (ii) Do Not Smoke within Designated Hazardous Areas;
 - (iii) Apply Lock-out Tag-out where isolation of Energy is required;
 - (iv) Do Not set off Naked lights within Designated Hazardous Areas;
 - (v) Do Not Work whilst under the Influence of Alcohol;

- (vi) Do not exceed the speed limit; and
- (vii) Do not work without fall prevention/protection in place.

G2.23.5 Factory Registration

- (a) Within four (4) weeks of Commencement Date, the Contractor shall register their works as a Factory. A copy of the licence shall be submitted to the S.O..
- (b) The Contractor shall be responsible for all safety requirements under the Contract, the Factory license, and applicable legislative requirements until completion of the Contract including any extensions.

G2.23.6 Safety Kick-off Meeting

- (a) Upon award of Contract, the Contractor's senior management shall attend a safety kick-off meeting conducted by the S.O. and attended by the Board to define and reinforce the Board's safety requirements and expectations, such as setting of safety targets, submission schedules, report formats for safety observations, safety inspections, accident statistics and near-miss incidents, reporting procedures for safety incidents and injuries, etc.

G2.23.7 HSE Management System (HSEMS)

- (a) Within 30 days of Commencement Date, the Contractor shall submit its proposed HSEMS in accordance to "SS 506, SS CP79", Code of Practice for Safety Management System for the construction worksites", and where applicable in compliance with ISO 14001 and OHSAS 18001 for the S.O.'s approval. The HSEMS shall incorporate all relevant legal and contractual requirements.
- (b) The HSEMS shall include the Contractor's company safety policy, safe work practices, risk assessments, in-house safety rules and regulations, small group activities, safety cycle activities, safety promotion programme, safety training, safety inspections, maintenance regime, subcontractor evaluation and selection, hazardous substances control, incident investigation, emergency procedures, occupational health programme and other such systems as per SS 506 and also SS CP 79: Code of Practice for Safety Management System for Construction Worksites. The HSEMS shall identify the relevant safety equipment and describe the safe method of work to be employed for each stage of construction. The HSEMS shall incorporate the requirements of this specification.
- (c) The HSEMS shall be displayed near the entrance of the Contractor's site office and elsewhere on site to be accessible to all workers. The Contractor shall display safety posters and large banners at the its site office and at its site exit/entry points.

G2.23.8 Responsibility

- (a) The Contractor shall be responsible for the health and safety of all operations in connection with the Contract and shall take all necessary actions to ensure the safety and welfare of all persons who may be on the site or adjacent thereto and at risk from activities on the site.
- (b) All persons working on the site shall be properly trained so that they are able to carry out their tasks and duties safely and in a manner that will not endanger the health and safety of them or others.

- (c) The Contractor shall be responsible for ensuring that his sub- contractors, any other contractors working on or interfacing with Contractor's site (interfacing contractors), and all persons entitled to be on the site comply with all relevant legal and contractual requirements including the Contractor's HSEMS and shall enforce its compliance.
- (d) If the Contractor is working within the site (factory) of another contractor, the Contractor shall comply with the other contractor's HSEMS or applicable safety requirements.
- (e) The Contractor shall ensure all persons working under its responsibility shall be issued with PPE with corporate logo clearly indicated front and rear of uniform and helmet. All such persons shall also display a valid sticker indicating completion of the required site safety induction courses which shall include both the overall Site induction course and course specific to the Contractor's own site.

G2.23.9 Safety Enforcement

- (a) The S.O. may require the removal from the Site of any person who fails to observe safety procedures, performing an unsafe act or promoting an unsafe working condition and that person will be barred from the project with immediate effect.
- (b) In the event that the Contractor fails to comply with the requirements of the applicable legislative requirements or Specifications, the S.O. will have the authority to instruct the Contractor to stop work in any particular location until the unsafe conditions or practices have been rectified to the satisfaction of the S.O.. In this event, the Contractor shall be responsible for all costs and delays resulting from such instruction and from complying with such instruction.

G2.23.10 Monthly HSE Report

- (a) The Contractor shall prepare a monthly HSE report and submit to the S.O. within five (5) days after the previous month completion.
- (b) The content of the monthly report shall include, but not limited to, the list set out below:
 - (i) Health & Safety Statistics (Month and cumulative summaries to be presented in graphical form);
 - (ii) Manpower Reports including man-hours worked;
 - (iii) Analysis of incident statistics: Accident Frequency Rates (AFR), Accident Severity Rate (ASR), Days Away from Work, Restricted or Transferred (DART);
 - (iv) Safety Incidents: Nos, Type, All Recordable/ Reportable (both human and non-human related, property and utility damage, dangerous occurrences);
 - (v) Observations, Near-Miss Reports and Lessons Learned shared;
 - (vi) Internal & External Audits - Finding Statistics (Observations, CARs, NCRs open, aging);
 - (vii) Emergency drills;
 - (viii) Evaluations and audits of subcontractor with actions taken;

- (ix) Training (include numbers attended): site safety induction course (SIC), other training and briefing sessions conducted;
- (x) Safety initiatives: Celebrations, Awards, Promotions, Wellness Programmes; and
- (xi) Environmental
 - 1) Incidents (Nos, Type);
 - 2) Environmental non-compliances received with details and actions taken;
 - 3) Status of following environmental management:
 - 3.1 Noise
 - 3.2 Vector Control
 - 3.3 Air Pollution
 - 3.4 Construction Waste Management
 - 3.5 Water Discharge
 - 4) Summary of Environmental Inspections;
 - 5) Security status:
 - Incidents (Nos, type)
 - 6) Inspection & visits by Agencies; and
 - 7) Public and External Complaints, Feedback, Actions Taken.

G2.23.11 Health, Safety and Environmental (HSE) Personnel

- (a) HSE personnel means designated personnel holding the following titles with appropriate competencies and certificates; **HSE Manager, Workplace Safety and Health Officer (WSHO)** registered with the Ministry of Manpower (MOM), **Earth Control Measures Officer (ECMO)** registered with the Institute of Engineers Singapore, **Environmental Control Officer (ECO)** registered with the National Environment Agency (NEA), **Workplace Safety and Health Coordinator (WSHC)**, **Workplace Safety and Health Supervisor (WSHS)** and HSE Administrative Assistant.
- (b) The ECO and WSHO shall each have five (5) years of traceable work experience on a contract scope similar or equivalent in plant construction or public infrastructure projects. The Contractor shall designate a HSE Manager to lead HSE in his project. The Contractor shall ensure his HSE Manager has minimum seven (7) years of traceable HSE Manager work experience.
- (c) The WSHC and WSHS shall each have three (3) years of traceable work experience on a contract scope similar or equivalent in plant construction or public infrastructure projects.
- (d) For clarity, designation of a WSHO requires five (5) years' experience as WSHO and not to take account of his/her experience as WSHC or WSHS or ECO or ECMO. This applies respectively to all HSE personnel's work experience as per appointment listed in Section (g) below with exception of HSE Administrative Assistant's work experience.
- (e) Prior to their appointment, the Contractor shall submit the HSE personnel's resume with detailed listing of their past experiences with valid references for the S.O.'s approval.

- (f) The HSE personnel shall not be removed or replaced without the S.O.'s approval. The HSE personnel shall attend the works during normal working hours and shall be available by mobile contact during other hours when works are progressing. The S.O. shall require the replacement of the appointed HSE personnel if the performance of the HSE personnel is not up to the S.O.'s expectations.
- (g) The Contractor shall provide, as a minimum, the following full time competent HSE personnel for implementing effective HSE programs and risk management at the worksites:
 - (i) One (1) HSE Manager;
 - (ii) Two (2) WSHO;
 - (iii) One (1) WSHC per shift;
 - (iv) One (1) WSHS;
 - (v) One (1) ECO;
 - (vi) One (1) ECMO; and
 - (vii) One (1) HSE Administrative Assistant.
- (h) The above clause represents the minimum expectations; however, the Contractor is responsible for adequately planning and resourcing for the project to ensure HSE standards are met. If deemed necessary by the S.O. the Contractor shall appoint additional HSE personnel to ensure adequate health and safety coverage for all Contract related works whenever work is in progress. The Contractor shall be responsible for the additional cost.
- (i) The Contractor shall appoint the WSHO, ECMO and ECO within one (1) month of Factory Registration or when any work commences on site, whichever is earlier, and subsequent WSHO no later than three (3) months thereafter or when directed to by the S.O..
- (j) The Contractor shall provide appropriate and qualified covering officers for WSHO, ECMO and/or ECO during their periods of absence due to annual leave, sick leave, National Service and trainings etc.
- (k) In addition to duties as prescribed in the Workplace Safety and Health (Construction) Regulations 2007 for WSHC and Workplace Safety and Health (Workplace Safety and Health Officers) Regulations 2007 for WSHO and National Environment Agency for ECO and ECO Coordinator, the HSE personnel shall carry out the following:
 - (i) Conducts regular safety observations and inspections;
 - (ii) Record, collate and submit the safety observation reports, monthly site inspection reports, accident statistics reports and near-miss report to the Board and the S.O.;
 - (iii) Provide the necessary safety training and or refresher courses to the full-time competent qualified supervisors; and
 - (iv) Comply with all relevant requirements in this Specification.

G2.23.12 Subcontractors

- (a) The Contractor is responsible for adequately planning and resourcing the Contract to ensure health and safety standards are met and maintain effective safety supervision at site during all work hours.
- (b) Every sub-contractor to the Contractor, at whatever level (sub, sub-sub, etc.) who employs more than 20 persons to carry out work at the site, shall appoint at least one full time supervisor.

G2.23.13 Method Statement and Risk Assessment (MSRA) and Design for Safety (DfS)

- (a) The Contractor shall identify all safety critical activities and ensure that a method statement is prepared for each activity and accepted by the S.O. before commencement of such activities. Risk Assessments conforming to the Singapore WSH Council's risk assessment guidelines shall be submitted with all method statements. Briefings to the S.O. are to be conducted for medium to high-risk activities.
- (b) The Contractor shall address all comments on the method statement and risk assessment arising from the S.O. review. The Contractor shall fully comply with the method statement approved by the S.O.. If there is any intention to change the method of work, the Contractor shall seek approval from the S.O..
- (c) The Contractor shall maintain a record of risk assessments conducted, including any control measures taken or to be taken and any safe work procedures.
- (d) The Contractor shall ensure all risk assessments and method statements are adequately briefed in relevant languages and disseminated to all relevant personnel prior to the particular work commencement. All persons involved in the work activity shall be briefed on the nature of the risk involved, the measures taken to minimise the risk, and the safe work procedures to control the risk.
- (e) The Contractor shall propose and engage a qualified and competent Risk Management Professional (RMP) and DfS Professional. The qualifications of DfS Professional proposed for this contract shall include:
 - (i) Have reasonable experience with safety and health for construction especially on infrastructure projects of a similar nature to this Contract;
 - (ii) Attended the DfS for Professional Course and passed the assessment, or equivalent;
 - (iii) Be a registered PE or an Architect with a Practicing Certificate or have 10 years relevant experience in the design (at least 5 years in design which includes contributions to designs, writing specifications) and the supervision of the construction of structures; and
 - (iv) Have a degree accepted by the Professional Engineer Board or Board of Architects and construction related degree accepted by Singapore Institute of Surveyors and Valuers, and Society of Project Managers.
- (f) RMP should be a HSE manager with relevant work experience in risk management.
- (g) The final approval for the appointed RMP and DfS Professional shall be within the Board's discretion.

- (h) The RMP and DfS Professional shall facilitate the hazard identification and risk management sessions. The Board and S.O. shall be invited to attend these sessions as observers.
- (i) The RMP and DfS Professional shall be the appointed person in-charge of the preparation and submission of the risk assessments. The hazard registers are live documents. The RMP and Dfs shall add new hazards that are identified, monitor and update the hazards in the register to ensure all hazards are mitigated and closed.
- (j) The RMP and DfS Professional shall attend the S.O. Project Safety Committee Meetings and any other risk management meetings directed by the S.O..
- (k) For all works to be designed by the Contractor, including temporary works, the Contractor shall perform Design for Safety (DfS) in compliance with the WSH Design for Safety regulations. The DfS shall be conducted by Engineers who have attained the Design for Safety Certificate.
- (l) The DfS shall be completed sufficiently in advance of the works commencing to incorporate the results of the DfS into the design of the works.
- (m) The Contractor shall invite the S.O. to all DfS workshops.

G2.23.14 Permit To Work (PTW)

- (a) The Contractor shall prepare, implement and comply with Permits to Work (PTW) for all work as required by legislation or by the Board or S.O.. The S.O. shall audit the Contractor's PTW to verify compliance.
- (b) The S.O. shall implement a centralised PTW system for the mandatory use of all contractors working on TWRP. The system will be a commercially available product with the objectives to streamline the preparation of PTWs, provide consistent standards and content for PTWs for all TWRP contracts, and to document, report, and manage all PTWs in a centralised system. The S.O. will administer this system and provide access for Contractor to utilise the PTW system. The PTW system is not intended to relieve the Contractor of his statutory obligations for implementing a permit to work system and his duties as occupier of a worksite. The Contractor shall be given the opportunity to make revisions to the permit to work forms that he deems necessary to maintain compliance with his obligations and shall be required to certify that the permit to work forms satisfy the Contractor's statutory obligations.
- (c) The PTW shall be valid only for the day or shift unless otherwise agreed upon by the S.O..
- (d) The safety assessor and the occupier's project manager approving the PTW shall be separate persons. In addition, the safety assessor shall be occupier's competent supervisory staff.
- (e) In the event that the PTW system by the S.O. is not available, the Contractor shall utilise its own permit to work system until the S.O. PTW system is available.

G2.23.15 Safety Training

- (a) The Contractor and its HSE representatives shall attend necessary safety training under the statutory regulations and requirements, including training mandated by the Workplace Safety and Health Act and regulations applicable to its employees and subcontractors, or as directed by the S.O.. The Contractor shall keep records of all training conducted and submit the same to the Board and S. O. when so requested.

- (b) The Contractor shall ensure that all of its staff, workers and subcontractors on site attend the mandatory Site safety induction course for the Site to be delivered by the Board's Site Development Works Contractor (C1A) under supervision of the S.O.. This Site safety induction course will include participation in the Safety Training Facility which will provide simulated experience with key high risk hazards expected for the site. None of the Contractor's staff, workers and subcontractors shall be allowed to work on Site without completion of all Site safety induction courses required by S.O..
- (c) The Contractor shall also develop and conduct a safety induction for all his staff and workers, subcontractors, Qualified Person (QP) and the Board's and S.O.'s staff for the Works to be performed by the Contractor. No one shall be allowed on the site(s) of the Contractor until they have attended the Contractor's mandatory safety induction course. The Contractor shall keep records of the participants and the induction details and submit the same to the Board and S.O. when so requested.
- (d) The Contractor shall provide a training room at its own site capable of providing the site induction course and other training to at least 20 workers at a time. The training room shall be provided with all the necessary audio, furniture and visual training facilities.
- (e) The Contractor shall ensure that no personnel work on the Contractor's Site prior to the completion of the Contractor's safety induction training and obtaining a security pass. The Contractor shall ensure that training information is translated where necessary in languages understood by the trainees. In addition, the Contractor shall have a system to clearly identify employees who have not worked in Singapore, during their first sixty (60) days on site in order to provide closer observation, such as buddy system with experienced worker, in order to more closely supervise their work to verify compliance with Project health and safety requirements.
- (f) The Contractor shall employ qualified operators for all the machinery to be used on Site even if it is not required by legislation. The operators shall possess a Skills Evaluation Certificate (SEC) from the Building Construction Board Academy or other MOM approved training centre. For machinery where there is no skilled training available in Singapore for example asphalt roller operator's course, the Contractor shall engage the supplier of the machinery to train them and authorize them in writing. Examples of qualified operator include, but not limited to: gantry crane operator, excavator operator, boring/piling operators, boom lift operator and welders.
- (g) The Contractor shall ensure that all his supervisory staff have attained the Building Construction Safety Supervisors (BCSS) Certificate or Equivalent. In addition, construction/project managers are to have attained Construction Safety course for project managers.
- (h) The Contractor shall develop and implement a comprehensive training and competency assessment system to assure the competency of his supervisory staff, lifting crew and machine operators prior to their deployment for works. The assessment system shall include face-to-face interviews and written tests that adequately evaluate appreciation of safety hazards associated with respective works, Safe Work Practices etc.

- (i) The Contractor shall ensure that all personnel and in particular, new personnel, or personnel transferred to new assignments are given proper safety training relevant to their duties.
- (j) The Contractor shall implement an identification system on Site to clearly identify all the qualified personnel and operators. This information shall be included with each person's personnel information included within the Site access control system.

G2.23.16 HSE Coordination Meeting

- (a) The Contractor shall conduct weekly health and safety co-ordination meetings with his sub-contractors and interfacing contractors to ensure that works are carried out on Site with minimum risk to workers and to the public. The meeting shall review all works on site including the movement of plant, equipment and hazardous materials, review, safe work practices, permit to work procedures, training, PPE and incidents. The meeting shall also inform personnel of potentially dangerous works operation and key risks at the site.
- (b) During progress coordination meetings on Combined Services Drawings (CSD), Structural, Electrical and Mechanical (SEM) and similar type meetings and reviews, the following items shall be included in the meeting agenda:
 - (i) Relevant lesson learned case study;
 - (ii) Planning and sequencing of work activities and identify incompatible works between contractors working in the same area;
 - (iii) Identification of risks and hazards of the interfacing works, including the conduct of site walk to verify these hazards;
 - (iv) Highlighting potential high-risk zones during handing over; and
 - (v) Developing a site map to show delivery routes and designated storage area for the Contractor and Interfacing Contractors.

G2.23.17 HSE Committee

- (a) The Contractor shall establish a health safety and environment committee regardless of the number of workers. The committee shall comprise management, site supervisor staff, safety and environment representatives from the Contractor and his sub-contractors including any interfacing contractors. The meeting chairman shall be the Contractor's Project Director or Project Manager. The meeting secretariat shall be the Contractor's Safety Officer or Safety Manager. The S.O. staff shall be invited to sit in the committee on an ex-officio basis.
- (b) The committee shall inspect the site on the same day before each month's committee meeting, or not later than one week before each month's meeting.
- (c) The Contractor shall include the following topics at a minimum on the meeting agenda:
 - (i) Sharing of objectives, strategies, safety mission statement and Life Saving Rules;
 - (ii) Chairman's review of HSE performance/ condition;
 - (iii) Report from HSE representatives;

- (iv) HSE Committee inspection report;
 - (v) Accidents and incidents with lessons learned and implementing actions;
 - (vi) Status of key HSE initiatives and programs;
 - (vii) Major concerns and actions to address; and
 - (viii) Authority visits, and discussion of follow up actions.
- (d) The Contractor shall ensure that all major decisions and actions made at each meeting are effectively communicated for implementation. The Contractor shall ensure that all major decisions made, follow-up actions required, and lessons learnt from incident reviews at each meeting are effectively communicated for implementation. Records of closed items arising from meeting shall be kept in records and submit to the Board and S.O. when so requested.

G2.23.18 Tool Box Meetings / Pre-Task Talk

- (a) Tool Box meetings shall be conducted twice daily specific to the work performed for the day; in the morning before work commences and right after lunch. Workers shall be briefed on the day's activities, the HSE precautions to be observed, the safe work practices to be followed. Each individual's PPE will be checked as to its suitability, and its correct use explained where necessary. If night works or multiple shift works are performed, then tool box meetings shall be conducted before shift work commences and mid shift break.
- (b) The outcomes from the HSE committee meetings shall be cascaded down to the workers through the Tool Box meetings.
- (c) The lessons learnt from incidents that occurred at the Site and any other relevant safety matters shall also be made known by the Contractor to the workers and other relevant persons at the tool box meetings.
- (d) The Contractor shall provide interpreters, where necessary, to ensure that the workers and other relevant persons of various nationalities understand the information being disseminated.
- (e) The Contractor shall plan daily Pre-Task Talk sessions led by the relevant Task Supervisors or Engineers to discuss the specific task methods to ensure every worker member is aware of the task risks, understands the method statement(s), and understands the necessary safe working methods.

G2.23.19 S.O. Project Leadership Forum

- (a) The S.O. will establish a TWRP site-wide HSE Leadership forum to meet on a regular basis (e.g. quarterly or monthly) to discuss site wide HSE goals, issues, incidents, lessons learned and other matters. The Contractor's senior project management and senior HSE personnel for the project shall participate in these meetings. Further details for these meetings will be provided by the S.O..

G2.23.20 Accidents and Incidents Reporting

- (a) In addition to the statutory and regulatory reporting requirements and the insurance requirements, the Contractor shall notify the S.O. of each accident, incident, dangerous occurrence or near miss concerning health, safety and environment. This shall include first aid injury even if they do not result in lost time. Verbal notification to the S.O. and notification via mobile app to the Board's and S.O.'s leadership immediately within the first hour, with further updates every four (4) hours after the incident and an initial written incident investigation report within 24 hours. Final conclusive written incident investigation report shall be submitted to the S.O. and the Board within five (5) working days. The Contractor shall put together a thorough presentation of the incident to the S.O. and the Board within fourteen (14) calendar days unless extended by S.O. due to unforeseeable circumstances.
- (b) Failure to provide immediate notification to the S.O. may negatively impact the Contractor's assessment under the Monthly Health, Safety and Environmental Incentive and Disincentive Scheme.
- (c) The Contractor shall propose remedial measures to prevent recurrence of the accidents and incidents to the satisfaction of the S.O..
- (d) The Contractor shall investigate each incident and submit an investigative report that defines what happened, why, direct and indirect causation factors, corrective actions and any other relevant learning. The report shall include relevant photos, sketches and evidences related to the incident or accident in soft and hard copies as deemed necessary to the satisfaction of the S.O..
- (e) The Contractor shall comply with all statutory and regulatory accident and incident reporting requirements. All information reported shall be copied to the S.O. at the time of reporting to the relevant authority or authorities. The Board's incident reporting process is set forth in the Appendices and the Contractor shall comply with its role in this process. The Contractor shall ensure that all incidents and accidents that result in any property/utility damage, environmental incidents, fatal injury, injury requiring first aider or Doctor's medical attention, hospitalization or other serious personal injury and all fires are recorded and fully investigated to the satisfaction of the S.O., although the mandatory requirements by Singapore Ministry of Manpower only stipulates that any work-related injury which renders and employee unfit for work for more than three (3) days must be reported.
- (f) The S.O. will establish a mode of anonymous communication for staff and workers to report safety lapses, concerns, unsafe conditions and acts. This will be advised to the Contractor after award for communication to its staff and workers.

G2.23.21 Emergency Response Plan

- (a) The Contractor shall establish an emergency response plan to respond effectively to emergency situations on its site. The plan shall be submitted to the S.O. for approval prior to any construction activities and updated at least on a quarterly basis.

- (b) The plan which includes, but not limiting to the following shall be communicated to all personnel within the site:
 - (i) Emergency response procedures, including rescue operations;
 - (ii) Evacuation procedures;
 - (iii) Medical emergency response procedures;
 - (iv) Composition of the Contractor's emergency management team including their roles and responsibilities;
 - (v) Coordination with the Board and the S.O. including Site-wide emergency response plans and the crisis management centre to be managed by the Board and S.O.;
 - (vi) Notification of relevant parties such as the Board, S.O., Ministry of Manpower (MOM), Singapore Police Force (SPF) and SCDF; and
 - (vii) Crisis communications protocols with other contractors, public and the media.
- (c) Emergency exercises and drills shall be conducted on a quarterly basis. Emergency table top exercises and drills with SCDF and all relevant agencies shall be on a half yearly. In-house exercises shall be conducted in quarters where exercises with SCDF are not being conducted. In-house drills shall be conducted in quarters where drills with SCDF are not being conducted. The types of emergency exercises to be held include the following, where applicable to the scope of the works:
- (d) In-house emergency exercises and drills shall be conducted on a quarterly basis. Emergency table top exercises with SCDF and all relevant agencies shall be on a half yearly basis while drills shall be at least once a year. The types of emergency exercises to be held include the following, where applicable to the scope of the works:
 - (i) Fire and Explosion;
 - (ii) Tower Crane operator rescue from height;
 - (iii) Work at height rescue;
 - (iv) Confined space rescue;
 - (v) Ground/Tunnel cave in rescue;
 - (vi) Disease outbreaks i.e. Dengue, Zika, SARS; and
 - (vii) Flammable chemical spill containment.
- (e) The emergency preparedness plan shall also include environmental pollution scenarios such as spillages of unauthorized/pollutive materials into sewage, watercourses or land and security/terrorism related scenarios.
- (f) In the event of any emergency on the Site, the Contractor shall immediately inform the relevant authorities and the S.O. and keep the S.O. informed throughout the emergency response. The Contractor shall follow any instructions of the S.O.. The S.O. will have an emergency response plan in place for the entire TVB Site.

G2.23.22 Use and Selection of Personal Protective Equipment (PPE)

- (a) The Contractor shall provide, maintain and enforce the usage of PPE for all its personnel on site always. The following PPE shall be compulsory on site and shall conform to the standards defined in the Appendices. The Contractor shall submit its proposed PPE to the S.O. for approval:
 - (i) Safety helmets with chin strap and ratchet suspension system conforming to SS98;
 - (ii) Safety hand gloves complying with EN 388 cut resist grade 4543 or Level 5. Gloves for specific tasks, such as chemical handling, electrical works, hot works shall comply with relevant standards such as EN 374 (chemical handling), EN 60903 (electrical – depending upon voltage) and EN 12477-Type B (hot works);
 - (iii) The Safety footwear shall be minimum ankle high boots, crack resistant sole, Zipper or Elastic or Lace upper, slip, oil and chemical resistant, steel toe cap, pierce resistant and Anti-static. Shall conform with SS 513 and AS/NZS 2210;
 - (iv) Every person required to work in water, wet concrete or other liquid shall be provided with suitable waterproof steel toe cap and pierce resistant boots compliant to SS 513;
 - (v) High-visibility reflective vest and uniform conforming to BS EN 471, Class 2;
 - (vi) Industrial cool lightweight and ventilated High-visibility long sleeved uniform (Top/Bottom) conforming to any combined or individual standards of (AS/NZS 1906.4, 4602.1, 2919, 4399, 1020), (EN 340, 1149, 13034, 471) and Sun Protection SPF 40 or 50 coated;
 - (vii) Where necessary, Heat and flame resist long sleeved uniform (Top/Bottom) conforming to any combined or individual standards of (NFPA 2112, 70E Category 2), (EN ISO 11612, 14116), (ASTM F1506, D6413), (AS/NZS 1906.4, 4602.1, 2919, 4399, 1020), (EN 340, 1149, 13034, 471). Flame Rating logo must be clearly visible;
 - (viii) Hearing protection with its container or Disposable ear plugs conforming to SS 549 with Noise Reduction Rating (NRR) of minimum 25 Decibels;
 - (ix) Eye protection glasses/ goggles conforming to SS 473 and ANSI Z87.1;
 - (x) Safety belts shall not be used or provided for restraining falls. Instead safety harness shall be used for fall protection. Safety harness shall comply with SS528 series (Personal fall-arrest systems) and SS570 (Personal protective equipment for protection against falls from a height – Single point anchor devices and flexible horizontal lifeline systems) or ANSI Z359. Lanyards shall not exceed 1 m in total length including unexpended shock absorbing block;
 - (xi) Personals working above 6 m of fall height shall be issued with a pair of suspension trauma stickers attached onto their harness;
 - (xii) Respirators / dust masks of the appropriate standard shall be provided for activities generating dust or fume;
 - (xiii) Every worker required to use or handle alkaline, acid or other corrosive substances shall be provided with suitable protective clothing and PPE;

- (xiv) The Contractor shall maintain and update all PPE issuance records. Additional PPE shall be issued to personals based on the type of work risk identified i.e. Face shield, Cartridge respirator, N95 haze mitigation, welding related PPE, working with chemicals PPE etc; and
- (xv) All PPE shall have clearly indicated/stamped manufacturer's month and year. The Contractor shall take into account from the manufacturer's month and year of any PPE to be replaced every two (2) years or sooner depending on its wear and tear condition. PPE brand selection for the S.O. and the Board shall be presented before procurement.
- (b) The Contractor shall comply with the standardization of helmet and vest colours for workers on DTSS2 project sites. The vest can be substituted for Hi visibility uniform of the same color:
 - (i) Blue helmet – Safety personnel / ECO;
 - (ii) White helmet – Supervisor/ Foreman/ Engineer/ Management;
 - (iii) Red helmet and high visibility two-piece luminous yellow uniform with title: 'Traffic Controller';
 - (iv) Red helmet and high visibility two-piece luminous yellow uniform: 'Banksman';
 - (v) Red helmet and red reflective vest with title: 'Rigger/Signalman';
 - (vi) Brown helmet and green reflective vest with title: 'Lifting Supervisor';
 - (vii) Green helmet and green elastic reflective vest with title: 'Welder'; and
 - (viii) Yellow helmet– General Workers.

G2.23.23 HSE Promotion

- (a) The Contractor shall develop an annual HSE promotional programme to demonstrate his commitment to advancing the HSE culture at the Site and reinforcing the concept that HSE and construction are inseparable. The programme shall enhance personal HSE awareness and influence all attitudes and behaviour of all personnel on HSE matters. The programme shall consist of general promotional activities which are carried out as part of a day-to-day activity and high impact promotion activities which are carried out as a campaign to reinforce a particular HSE point at the Site. The HSE promotional programme shall be revised and updated at least once a year.
- (b) The Contractor shall organise a minimum of two (2) campaigns covering health, safety and environmental related topics for each calendar year.
- (c) The Contractor shall also continually work to raise safety awareness through celebrations and competitions such as slogan competitions, poster designs, to be held throughout the course of the project. The Board and S.O. shall support and attend the programs organised to witness and demonstrate support for the activities. Safety promotion activities organised must at a minimum meet the following requirements:
 - (i) Proposed event to be monthly or half yearly;
 - (ii) Budget allocation inclusive of awards, prizes, meals and refreshment;

- (iii) Specific theme that is relevant to the next phase of work, work currently being performed or specific to safety areas needing improvement;
 - (iv) A committee to be set up to judge competition entries and the winning entrants must be recognised and receive a prize;
 - (v) The Contractor's workers and employees should participate to enter slogans, poster drawings or ideas to improve safety and to raise the safety awareness;
 - (vi) The winning posters, slogans, t-shirt, etc., shall be published and distributed to all work areas; and
 - (vii) Winning ideas shall be implemented and awarded accordingly.
- (d) Contractors shall organise regular safety promotions and awareness celebrations annually and for the overall project, inclusive of the expenditure for meals, refreshments and the prizes and awards.
- (e) The Contractor shall conduct regular 'Safety Time-out' sessions of at least one per month. This is to allow the Contractor to take stock and refocus on safety, review current work activities and its associated hazards, as well as to identify additional safety measures required to maintain high standards on the worksites. The Contractor may conduct the 'Safety Time-out' sessions on a site-wide basis or to focus on specific work activities or subcontractors at different stages of their works. The Contractor shall submit a schedule and programme for the 'Safety Time-out' to the S.O. for approval.
- (f) The Contractor shall divide the worksites into designated work areas, each led by his competent supervisors who shall be responsible for the area's HSE performance. The HSE performance for each work area shall be assessed monthly and corrective actions shall be taken to raise safety performance. The monthly HSE assessment shall include, but not be limited to accident statistics and substandard practices and conditions recorded at various HSE inspections for the particular work area. The teams that meet HSE targets shall be duly recognised and rewarded. The recognitions shall include both monetary and non-monetary rewards and be given out at a suitable event attended by the workforce or as directed by the S.O..
- (g) The Contractor shall provide, erect, maintain and finally remove when ordered, an Accident Statistics Board (ASB) 3m x 2m in size written boldly in English, the content of which shall include, but not limited to the following: Contract start date, date of update, total man hours worked; total number of recordable incidents; total lost-time accidents; total fatalities; hours worked since last incident/accident, total crane collapses; frequency rate; and severity rate.
- (h) The ASB shall be erected in a prominent location, preferably near the main entrance to the Site, which shall be approved by the S.O.. The Contractor shall be responsible for ensuring that the statistics are updated daily to reflect the status of the HSE performance at the Site.

- (i) The Contractor shall produce at least two health and safety digital videos yearly to raise the health and safety awareness and standards of its workforce. The topics and contents for the videos shall be agreed with the S.O. and shall be minimum 10 minutes in duration. The videos shall be professionally produced, be of High Definition (HD) quality and translated into the respective native languages of the workforce. The Board and S.O. shall be given reasonable use rights to the videos.

G2.23.24 Evaluation, Selection and Control of Sub-Contractors

- (a) The Contractor shall include legislative and Board's site-specific HSE requirements in tender packages for their sub-contractor selection and conduct pre-job meetings to address job HSE expectations before awarding them.
- (b) The Contractor shall select sub-contractors who have attained bizSAFE level 3 certification, awarded by the Workplace Safety and Health Council, prior to their work commencement at Site. Should there be sub-contractors who are not bizSAFE-certified, the Contractor shall ensure that such sub-contractors be certified to bizSAFE level 3 within a 6-month period, upon informing S.O. of its intention to engage them. Notwithstanding this, the responsibility remains with the Contractor for all of its sub-contractors to be bizSAFE level 3 certified as early as possible.
- (c) The S.O. may participate in sub-contractor pre-tender meetings, selection process, pre-job meeting reviewing the HSE performances. The S.O. reserves the right to reject the sub-contractors who are not certified within the timeframe.

G2.23.25 HSE Inspection

- (a) The Contractor shall carry out internal HSE inspections at least once a day or at least once per shift. In addition, informal spot checks should be carried out more frequently on critical site activities.
- (b) A written record shall be kept of the daily inspection findings and the results of inspections should be brought to the line manager having responsibility in the area concerned, together with the necessary remedial action and due date for completion. Any corrective action shall be immediately implemented by the line manager, and followed up by the HSE personnel. The Contractor shall submit records of inspection report as deemed necessary by the S.O..
- (c) Inspection of shoring of formwork, side supports of excavations and trenches, cranes and scaffolds should be carried out after any episode of inclement weather which may affect their stability or integrity.
- (d) The Contractor's senior site management shall participate in the S.O.'s weekly, monthly, quarterly, and annual ad hoc safety inspections. The Contractor shall close out the inspection findings to the full satisfaction of the S.O..
- (e) The S.O. shall require the Contractor to suspend a part of the works or the whole of the works if it is deemed to be unsafe. The Contractor shall be required to rectify the substandard condition(s) or practice(s) to the satisfaction of the S.O.. In such events, the Contractor shall not be entitled to any claim for compensation or Extension of Time for Completion.
- (f) The Environmental Control Officer (ECO) shall carry out daily inspections and submit findings and remedial actions with photographs to the S.O. fortnightly.

G2.23.26 Maintenance Regimes for all Construction Plant, Equipment, Vehicles and Tools

- (a) The Contractor shall assess the HSE risks especially in terms of age, noise, emissions, condition, etc. associated with the plant, equipment or tool and only those assessed with minimal HSE risks shall be brought to the Site.
- (b) The Contractor shall ensure that machineries such as cranes, excavators and other mobile rigs shall have guard rail barricades installed on its frame to prevent falling from height that exceeds 1.8 m when any person access the top of the machine for maintenance and inspection purposes. Access to flatbed trailer shall be safely provided by means of steps/ladder with handrails or equivalent to mitigate the risk of fall from height. Machineries not originally fitted with such guard rails and steps shall be retro fitted at the Contractor's own cost.
- (c) The S.O. shall stop the plant, equipment, vehicle or tool from entry or operation or require its removal if he finds the HSE associated risks to be critical. The Contractor shall not be entitled to any claim for compensation or Extension of Time for Completion.
- (d) The Contractor shall implement a preventive maintenance programme to ensure that all plant, equipment and tools are maintained in a safe and working order.
- (e) The Contractor shall implement a monthly inspection program to inspect all plant, equipment and tools with suitably qualified personnel. Contractor shall conduct a thorough inspection which shall bear a color-coded sticker or tag with the Contractor's name, equipment identification, e.g. serial number, manufacturer's model or serial number, date of recent inspection (or due date of the next inspection) and the signature of the competent person inspecting the equipment. Plant and equipment to be inspected shall include, but not necessarily be limited to, electrical tools, generators and installation assemblies, lifting machines, lifting gears, fire extinguishers, fall protection gear, construction mobile machines, compressors. All inspections should be documented.
- (f) All vehicles shall be duly inspected by the Contractor prior entry in the Project, that the vehicle is in good working conditions. The S.O may reject the vehicle if he finds the HSE associated risks to be critical. The Contractor shall not be entitled to any claim for compensation or Extension of Time for Completion. The following lists is a guidance:
 - (i) Brake is working;
 - (ii) Tyre with minimum three (3) mm tread depth;
 - (iii) Tyre pressure;
 - (iv) Tyre side wall damage; and
 - (v) No excessive emittance of white smoke or exhaust smoke.

- (g) Contractor to ensure all equipment without a valid inspection or valid colour coding shall not be used and must be removed from the Site with the access pass cancelled. Should the Contractor still need to use the equipment at site, the process of obtaining access for that equipment has to be re-initiated. The monthly colour codes shall be in different unique colours every month after inspections and with its unique colour:

MONTH	COLOUR
January, May & September	GREEN
February, June & October	WHITE
March, July & November	BLUE
April, August & December	YELLOW
Defective items / Due Servicing	RED

- (h) All plants, equipment and tools that have undergone repair or maintenance shall be inspected and checked before being returned to service. Stickers or tags shall be displayed to indicate its approval for usage or “Not for Use”.
- (i) The Contractor shall implement a lockout and tag-out system in accordance with SS571, Energy Lockout and Tagout Procedure. Plant and equipment such as machinery and vehicles of any types which can be started by a key shall be secured against incorrect or unauthorised restart at any time.
- (j) Job-made or modified tools of any kind shall not be used on site.
- (k) All rotating machines and tools, including but not limited to power saws, cutting wheels, grinders, portable pumps, shall have suitable protective guards on rotating parts and Deadman switch installed where applicable. Portable power operated hand saws shall be equipped with guards above the base plate which completely protects the operator from contact with the saw blade when in motion and with self-adjusting guards below the base plate which completely covers the saw to the depth of the teeth when the saw is removed from the cut.

G2.23.27 Hazardous Substances and Chemicals

- (a) The Contractor shall assess the Safety Data Sheets (SDS) of all the hazardous substances and chemicals prior to its entry to site for its suitability in terms of HSE hazards and consider safer alternatives.
- (b) The S.O. may require the removal of any hazardous substance or chemical if there are safer alternatives. In such events, the Contractor shall not be entitled to any claim for compensation or Extension of Time for Completion.
- (c) The Contractor shall ensure that all hazardous substance and chemical containers are labelled, its movement is recorded and returned to the designated storage areas when not in use.

G2.23.28 Monthly Health, Safety and Environmental Incentive and Disincentive Scheme

- (a) The Contractor is encouraged to aim for an Exemplary score in the Board's Monthly Health, Safety and Environmental (HSE) Performance Incentive and Disincentive Scheme for Contractors. The Board intends to offer incentives to the Contractor for good HSE practices but will also impose disincentive deductions from the Contract Sum for poor HSE performance.
- (b) The Contractor shall be subject to the Board's scheme of incentives and disincentives. Incentives will be provided by the Board to the Contractor over and above the base contract sum if HSE performance is exemplary by demonstrating high degree of commitment and demonstrable actions. The maximum incentive amount that the Contractor may earn during the Contract is defined in the Provisional Sums. However, if the Contractor is unable to meet the HSE performance criteria, the Board will impose disincentive deductions on the Contractor's Contract Sum for poor HSE performance of his own staff and his sub-contractors, including the inadequate supervision of the subcontractor's work and hence affecting the HSE performance on the Project. The maximum disincentive amount during the Contract shall not exceed the maximum incentive amount.
- (c) The Contractor shall ensure that his entire project team including its sub-contractors proactively takes all the necessary steps in ensuring exemplary HSE performance in the project. The Contractor's role and commitment to this shall be driven and led by his senior management associated with the Project.
- (d) The Contractor's HSE performance will be assessed monthly by the Board and S.O.. At the end of each four (4) months during the Assessment Period, the Contractor's cumulative assessments will determine whether the Contractor has earned the incentive amount, has not earned the incentive amount, or is subject to the disincentive amount. Incentive amounts not earned in a four-month period will not be carried over into subsequent assessments.
- (e) Assessment Criteria
 - (i) The Contractor and his team shall be assessed and scored on the following Key Performance Indicators (KPIs) as may be relevant and appropriate at the time, in support of the TWRP safety targets. These KPIs will be reviewed annually by the Board and S.O. taking into consideration the various phases of design and construction.
 - (ii) The Contractor shall be assessed based on the following three (3) main assessment criteria:
 - a) Incident Statistics and Reporting (50 points)
 - b) HSE Promotion (40 points)
 - c) Environmental Conformance (10 points)

The KPIs for each of the above assessment criteria are defined below These shall be subject to review and change by the Board and S.O. yearly.

The Contractor shall demonstrate the fulfilment of the above KPIs by keeping the appropriate evidences and records for the Board and S.O.'s audit and review.

(f) Assessment Period

- (i) A briefing of the scheme will be conducted during the kick-off meeting. The assessment period shall begin one (1) month after the commencement of work on site. The assessment period shall include any Extension of Time (excluding periods under Liquidated Damages) and end on the date of the Certificate of Substantial Completion (CSC) of the Contract.

(g) Scoring System and Incentive Payment / Disincentive Deduction

- (i) The Contractor shall be assessed on a monthly basis, where points will be awarded for the three main assessment criteria based upon the KPIs for each assessment criteria and sum totaled for each month. The monthly points scored will be reviewed by the S.O. and the Board in a monthly review meeting with the Contractor with any follow-up actions for improvement and target dates to be identified.
- (ii) The Contractor's HSE Rating for the four months will be based on the percentage of HSE scores earned monthly during the four-month period/ maximum possible HSE scores for the four months, and rated as follows to determine if incentives are offered or disincentives imposed for each four-month period:

Monthly HSE Score	HSE Performance Rating	Incentive/Disincentive Award
> 85%	Exemplary	Incentive paid
50% – 85%	Needs Improvement	None
< 50%	Unacceptable	Disincentive deducted

- (iii) For exemplary HSE performance rating during each four-month period as determined by the monthly HSE review against the KPIs, the Contractor shall receive an incentive sum from the Board over the course of the Contract period, including any Extension of Time (excluding Defects Liability Period and periods under Liquidated Damages). The annual amount payable shall be determined by the Board and S.O. at the beginning of each year including the amount available to be paid for each four-month assessment period.
- (iv) Conversely, if the Contractor fails to achieve at least 50% rating against the KPIs during each four-month period as determined by the monthly HSE review against the KPIs, the Board will impose a deduction disincentive on the Contractor which involves the forfeiture of a portion of his Contract Sum over the Contract period, including any Extension of Time (excluding Defects Liability Period and periods under Liquidated Damages). The maximum amount to be deducted each year shall be also be determined by the Board and S.O. at the beginning of each year.
- (v) In the event of an incident involving Fatality during a particular month, no incentive payout shall be offered for the following 12 months. The Scheme will continue with monthly assessments and the Contractor may still be subject to the disincentive amounts during the following 12 months.

- (vi) The above incentive sum shall be drawn from the Provisional Sum in the Contract, whereas the disincentive deduction shall be deducted from the payment due to the Contractor under the Contract, on the approval of the Board.
- (vii) If the Contractor obtains an unacceptable performance rating for the four-month period, the Contractor may be placed under surveillance which will entail higher frequency of inspections and audits of their worksite(s). The Contractor's senior management shall also commit to an action plan to improve the site HSE performance, which shall include regular meetings with the S.O. to address the required actions. The Contractor shall carry out all remedial actions promptly within a given duration before the following month's assessment audit.
- (viii) Failure to make improvements to the worksite and rectify the HSE issues according to the action plans may result in the issuance of stop work orders by the S.O.. Any delays or disruptions to the progress of work and any associated costs due to the Contractor being placed under surveillance to rectify any poor HSE issues shall be borne by the Contractor and shall not impact the Completion Date as agreed in the Contract. A stop work order shall only be revoked upon acceptance by the Board, S.O. that the worksite is in a safe condition.
- (h) Monthly HSE Assessment
 - (i) The audit team carrying out the monthly assessments at the worksite to assess the Contractor's HSE performance will comprise management and safety representatives from the S.O. and the Board. The Contractor shall participate in and allow access to the site and provide all necessary attendance and information to the audit team for the performance of such assessments.
 - (ii) The Contractor's senior management, Project Manager, HSE Manager or WSH Officer and the relevant sub-contractor representatives shall be present to provide the necessary attendance to the audit team during the monthly HSE assessments.
- (i) Monthly Assessment Criteria for HSE Key Performance Indicators:

No.	Monthly HSE Performance KPI	Max Score
1	Incident Statistics and Records	50
a	Accident Frequency Rate (AFR), Accident Severity Rate (ASR), Occupational Disease Incidence (ODI) incurred by the Contractor, including sub-consultants and subcontractors under his management and/or supervision.	30
b	Demerit Points incurred by the Contractor, including subcontractors under his management, premise and/or supervision.	5
c	Recordable Incidents, Lost Time Incidents and Man-days Lost incurred by the Contractor, including subcontractors under his management and/or supervision.	5

No.	Monthly HSE Performance KPI	Max Score
d	Property Damage Incidents, Environmental Incidents, Security Incidents, Dangerous Occurrences, Stop Work Orders related to HSE and Security, Life Saving Rules Infringements, and Contractual Specification Non-Conformances issued for Health, Safety, Environmental and Security incurred by the Contractor, including subcontractors under his management and/or supervision.	10
2	Health, Safety, Environment Promotion	40
a	Completion of weekly Project inspection including office space, with a minimum of 10 observations reported and recorded by Senior management* * Senior management defined as above Project Manager	10
b	Active drive towards the reporting of near-misses*, unsafe behaviors by member of the Contractor, including sub-consultants and subcontractors under his management or supervision. Minimum of 5 reports. Observations reported under 2a by Senior management will not be counted towards these criteria. * Near miss shall be defined per United States OSHA as “an incident in which no property was damaged and no personal injury was sustained, but where, given a slight shift in time or position, damage or injury easily could have occurred.”	5
c	Active participation and contribution by supervisors (Non HSE personals) of the Contractor in presenting HSE related trainings and safety moments. Minimum of 5 events.	5
d	Organize Weekly Project wide housekeeping stand down (2 hours) by Contractor that results in a high standard of housekeeping on the project site including access and roads within the Contractor’s site and work areas. The weekly stand down may be waived in the discretion of the S.O. should the Contractor maintain a high standard of housekeeping during the week.	4
e	Monthly inspection of all Heavy mobile equipment e.g. all Cranes (boom down), Piling Rigs, Excavators by a competent person/team.	4
f	Monthly project risk committee meeting to review upcoming key activities and identify risk mitigation actions that are communicated to the Contractor’s team.	2
g	Senior Management presence at each reporting and handling of incidents investigations, near misses, HSE audit findings followed by proactive sharing of lessons learned from such reporting and handling.	2

No.	Monthly HSE Performance KPI	Max Score
h	Regular Senior Management presence on site at a minimum of once a week for site walks, inspections, etc. and attendance in Toolbox Meetings.	3
i	Completion of a minimum of 12-hours of internal or external HSE training or refresher courses for each member of the Contractor's management, design and supervision team each year. Assessment will be made on a pro-rate basis to verify that Contractor is conducting regular training throughout the Assessment Period.	3
j	Cross project sharing on HSE including Best Practices with other Contractors involved in TWRP and tunnel contracts.	2
3	Environmental Conformance	10
a	Effective performance of the earth control measures (ECM) implemented with no reports of silt discharges into waterways or open sea.	2
b	Effective vector control (insects, rodents and other pests) implemented, e.g.: weekly fogging/oiling, clearing of possible mosquito breeding habitats, no reports of dengue, zika, etc.	2
c	i. Proper housekeeping and hygiene at designated canteen/dining areas and proper segregation and disposal of food waste; ii. Effective housekeeping practices for waste management, equipment and machinery storage and maintenance, water conservation programme. iii. Adequate provision and maintenance of sanitary and toilet facilities for workers	2
f	Effective protocols for handling, storage of hazardous substances, including regular reviews and training provided to the Contractor's team, e.g.; response to fires, spills and leaks.	2
g	Effective implementation of measures outlined in the Contractor's Noise Management Plan.	2

- (j) During the Contract Period, if the Contractor accumulates monthly incentive scores of less than 50% for four (4) consecutive months; or utilities damaged based on any of the criteria listed in table below, Contractor's top management will be called to explain the cause, provide effective recovery action plan to prevent recurrence, raise safety standards and reinforce their commitment to the Board and S.O..

(k) The following guidelines will be used to assess the KPI scores monthly. These guidelines will be reviewed by the S.O yearly.

(i) KPI 1 – Incident Statistics and Records

	KPI Criteria	Score
1	Complied with all four criteria (a-d)	50
1a	Within project AFR, ASR and ODI defined rates	30
1b	No Demerit Point issued for the month	5
	If Contractor is assessed demerit points for any of demerit point items 1-7, they shall not earn the entire 5 points. For demerit point 8, each one demerit point assessed in the month shall reduce the incentive points by 1 point. Refer to G2.23.28 – Demerit Points system.	<5
1c	If any recordable Incident (injury), LTI or man-days lost incurred in the month	0
1e	If any incidents of property damage or environmental or security; or any dangerous occurrences or stop work orders issued related to HSE and/or security; or any infringements of Life Saving Rules; or any Contractual Specifications non-conformances issued for HSE and/or security	0

(ii) KPI 2 – Health, Safety, Environment Promotion

	KPI Criteria	Score
2	Complied with all ten criteria	40
2a	Weekly contribution by Senior Management, 10 observations for the month	10
	Weekly contribution by Senior Management, 5 to 9 observations	5
	Weekly contribution by Senior Management, less than 5 observations	0
2b	Active drive towards reporting Near misses, Unsafe acts, Unsafe condition – 5 or more reports	5
	Active drive towards reporting Near misses, Unsafe acts, Unsafe condition – 3 or 4 reports	3
	Active drive towards reporting Near misses, Unsafe acts, Unsafe condition – less than 3 reports	0
2c	HSE related trainings and Safety moments conducted by supervisors – 5 or more trainings	5
	HSE related trainings and Safety moments conducted by supervisors – 3 or 4 trainings	3
	HSE related trainings and Safety moments conducted by supervisors – less than 3 trainings	0

	KPI Criteria	Score
2d	Weekly two hours Project wide housekeeping stand down conducted with documented report – 4 stand downs. Any weeks for which the S.O. has waived the requirement for the two hours stand down shall be counted as a completed stand down with report	4
	Weekly two hours Project wide stand down conducted with reports – less than 4 stand downs.	0
2e	Monthly inspection of all heavy mobile equipment, piling rigs, excavators – All active equipment are inspected	4
	Monthly inspection of all heavy mobile equipment, piling rigs, excavators – 50% or more equipment inspected	1
2f	Monthly Project Risk Committee meeting conducted (<i>Not Safety Committee meeting</i>)	2
2g	Senior Management presence at each incident investigation, near miss and safety audit report	2
	Senior Management presence at 50% or more of each incident investigation, near miss and safety audit	1
2h	Weekly involvement and presence by Senior Management in Toolbox talks, inspections – 4 times	3
	Weekly involvement and presence by Senior Management in Toolbox talks, inspections – 2 or 3 times	2
	Weekly involvement and presence by Senior Management in Toolbox talks, inspections – < 2 times	0
2i	Contractor Project team members including management, design, supervision staff excluding sub-consultants and sub-contractors complete specified hours of HSE training. External or internal HSE trainings will be acceptable with documentary evidence of training and attendance. .	3
2j	Monthly HSE Cross Project between Main Contractors sharing on Lessons learnt, Best practices, Site visit	2

(iii) KPI 3 – Environmental Promotion

	KPI Criteria	Score
3	Complied with all five criteria	10

G2.23.29 Demerit Point System

- (a) In addition to the HSES Performance Incentive and Disincentive scheme, the Board will also impose demerit points on the Contractor for every HSES non-compliance found during observations, inspections or any other form of audits on the worksite during the Contract period.

- (b) Demerit points shall also be imposed on any critical non-compliance which poses imminent danger to the worksite personnel, public or the environment that occurred in any worksite of the Contract during the Contract period. Examples of critical non-compliance include: working in a confined space without continuous gas monitoring as required under the WSH Act, performing works on high-tension voltage installations without authorisation and proper procedure of isolation, etc; carrying out hazardous work activities without a permit-to-work; working on uncertified/ untagged scaffolding; working at height without fall prevention plan and proper restraint/ arrest system; lifting operations not in compliance with WSH (Construction) Regulations, etc.
- (c) The demerit points system shall also be imposed on all recordable incidents, dangerous occurrences, occupational disease or environmental pollution that occurred in the worksite during the Contract.
- (d) The non-compliances noted above shall refer to breaches to relevant Legislation, Regulations, Orders, Singapore Standards, Codes of Practice, other standards or management systems recognised internationally and applicable in the Singapore context and Board's HSES requirements and the Contract Specifications.
- (e) The HSE Demerit Points System including Administrative Charges is as shown below:

No	Severity of Incident / Non-compliances Observed	HSES Demerit Points	Administrative Charges
1	Incident involving death or injuries to public or the death of a Contractor's or sub-contractor's worker	100	\$15,000.00
2	<u>Serious/ Life-Threatening Injuries to a Worker:</u> a) Amputation or permanent disability b) Hospitalised for 2 weeks or more c) Lost Time for 20 days or more	70	\$10,000.00
3	Dangerous Occurrence that could have resulted in Serious/ Life Threatening Injury to Worker(s)	50	\$4,000.00
4	<u>Non-Serious/ Non-Life-Threatening Injuries to a Worker:</u> a) All Lost Time occurrences of more than 3 but less than 20 days	20	\$4,000.00
5	Property / Utility Damage and Interruptions	20	\$2,000.00
6	Fines & Summons received	10	\$2,000.00
7	Critical non-compliance incident with imminent danger, e.g. relating to general public (motorists and pedestrians)	10	\$2,000.00

No	Severity of Incident / Non-compliances Observed	HSES Demerit Points	Administrative Charges
8	For every ten (10) non-compliances or safety lapses found	1	\$1,000.00

- (f) Worksites that have accumulated demerit points from categories 1-3 in the table above will be placed under surveillance with the Contractor committed to resolve all actions within a given duration to be agreed with the S.O..
- (g) All demerit points imposed on the Contractors for a particular month will be effective for a twelve (12) month rolling period.
- (h) An official warning letter will be served to the Contractor to inform them of the non-compliance and any relevant violated legislations or contractual requirements.
- (i) Administrative charges will be recoverable from the Contractor. The administrative charge is for investigative and administrative costs as well as expenses incurred by the S.O. and the Board as a result of the incidents or non-compliances observed and shall be deducted from any monies that are due or become due to the Contractor.

G2.23.30 Health, Safety and Environmental Audit

- (a) The Contractor shall appoint an independent MOM approved audit company to audit their HSE system in accordance with SS 506. The audits shall commence within two (2) months after factory registration by the Contractor and continue once every six (6) months after the initial audit. The audit company engaged by the Contractor shall be accepted by the S.O.. The audit company shall present and submit their audit methodology including detailed resumes of their audit team and audit checklist for the S.O. acceptance before commencement of any audits. The audit company is required to give a presentation of the audit findings to the S.O..
- (b) The audit report and its corrective actions shall be brought to the attention of all sub-contractors and copied to the S.O..
- (c) In particular, the audit shall also include auditing the environmental control practices, such as earth control measures, vector control practices and activities on site as specified in General Specification G9 Environmental.
- (d) Contractor shall submit a report on the findings and recommendations made during the cross-audit program to the S.O..
- (e) Deficiencies identified during S.O. and internal audits shall be corrected by the Contractor to the satisfaction of the S.O..

G2.23.31 Temporary Electrical Installations at the Site

- (a) The Contractor shall obtain licenses for using electrical power from their own generating sets.
- (b) All temporary electrical installations, equipment and tools shall be checked and certified safe for use prior to usage on site by a Licensed Electrical Worker (LEW) and thereafter monthly and after any repairs. The LEW shall provide a sticker on the equipment and tools indicating the date of inspection and is safe for use.

- (c) Apart from LEW certifying industrial tools, electrical installations and office appliances such as water dispenser, toaster, extension adapters, microwave ovens shall be certified by LEW on a quarterly basis. The Contractor shall ensure its sub-contractors comply. The Contractor shall implement a colour coding scheme for inspections in accordance with the plant and equipment colour coding sub-section of this Specification.
- (d) A current photograph of the LEW(s) and the contact number(s) shall be displayed on the outside of all boxes containing electrical distribution boards for ease of reference. These boxes must be secured with locks to prevent tampering and the keys kept with the LEW and Contractor's Safety Department.
- (e) The Contractor shall ensure that all portable electrical appliances used above and below ground level, that are hand held tools and inspection lamps, are rated at 110 V AC via a step-down transformer centre tapped to earth (CTE).
- (f) The Contractor shall ensure that all generators and welding sets in use on site are adequately and effectively earthed at all times during operation.

G2.23.32 Storage License for Petroleum and Flammable Materials

- (a) If the Contractor intends to store petroleum and flammable materials on site, it shall obtain a storage licence from SCDF and a copy of the licence shall be submitted to the S.O..
- (b) All diesel stored on site shall be kept in drums or in bulk tanks which in either case shall be located at a designated place away from any sources of ignition or open drain which does not lead to an interceptor and shall be properly labelled. A "No Smoking" sign shall be displayed at the storage location and a charged fire extinguisher of correct type kept on standby.
- (c) All bulk diesel tanks shall be properly supported in an elevated position to facilitate gravity discharge. They shall stand within a bund constructed to contain a volume of 110% of the volume of the tank. There shall be no breaches in the bund wall, no material shall be stored within the bund and rain water collecting in the bund shall be regularly removed to prevent build-up. The inner face of the bund wall shall be coated with a chemical resistant material. A chemical resistant valve, which shall be closed at all times, except for releasing rainwater into a storm water drain via an oil intercepting system, shall be installed at the outlet situated outside the bund, in accordance with the National Environment Agency (NEA) Code of Practice on Pollution Control.
- (d) All drums of diesel on site shall be in good condition and shall be kept closed with a lid or cap when not being used. They shall be stored on end with the lid or cap uppermost and kept within a tray of sufficient volume to contain the contents of the largest drum in the case of accidental rupture, taking into account the presence of other drums within the tray.
- (e) Drums of diesel shall not be rolled along the ground. They shall be transported vertically chained on a trolley or by a forklift fitted with a drum handling device and not standing unsupported on the forks or on a pallet or by crane using a safe slinging technique.

- (f) Diesel shall be transferred from the storage drum to another container, or to the tank of plant/machinery using a hand pump wherever practicable and a drip pan must be provided. Where the diesel container is light enough to be lifted by one person it can be poured out by hand a funnel must be used to guide the liquid.
- (g) Any spillages of diesel shall straight away be absorbed using sand or other absorbent materials, which shall be disposed of as contaminated waste. On no occasion should diesel be allowed to enter the site drainage system unless this is connected to an interceptor prior to the site waste being discharged into the public sewer system.

G2.23.33 Welding and Cutting

- (a) The Contractor shall comply with SS 510, Safety in welding and cutting (and other operation involving the use of heat).
- (b) When cylinders are used from a pallet, then a safety distance of 6 m shall be maintained between oxygen and acetylene cylinder pallets.
- (c) Minimum quantity of gas cylinders should be kept at work locations on site, and the remainder moved to the designated storage area at ground level. Cylinders should be secured in a vertical position and individual sets should be chained to trolleys or to a fixed support.
- (d) The Contractor shall implement a permit to work for all hot works to ensure that the equipment is safe from defects and no incompatible works are carried near the hot work area.
- (e) Liquid petroleum gas used below ground shall be subjected to the S.O.'s approval. Propane shall be used in a compressed air environment. Oxygen or acetylene cylinders taken underground shall be transported back above ground at the end of each working shift and stored in the designated storage areas.
- (f) Workers carrying out welding shall be provided with face shields compatible with safety helmets so that both can be worn at the same time.
- (g) Fire blankets shall be provided to contain sparks arising from welding and cutting operations.
- (h) Opaque screens shall be securely positioned around any electric arc welding being carried out on Site to protect other workers and passing members of the public, either on foot or as drivers or passengers in vehicles, from the arc. Such screens shall be maintained in good condition.
- (i) Cylinder valves shall be protected from damage by protection caps, valve guards or other effective means. Such protective means shall be in place whenever the gas cylinders are not in use or not connected for use.

G2.23.34 Lightning Warning Alert System

- (a) C1A contractor will install and operate a lightning warning alert system for the entire TVB site. Hours of operation of the system will be consistent with the working hours for the TWRP contractors. Information from this system will be communicated to all contractors for their use in managing their works and safety of all their workers in the event of forecasted or actual severe weather. The Contractor shall not have cause for any claim to the S.O. for the information provided by the S.O. from this system.

G2.23.35 Fire Safety Plan

- (a) The Contractor shall establish a Fire Safety Plan to ensure that the work on site is undertaken to the highest standard of fire safety. As a basic guide the Contractor shall address the requirements contained in the Joint Code of Practice on the Protection from Fire of Construction Sites and Buildings Undergoing Renovation - under the title "Fire Prevention on Construction Sites" published by the Construction Confederation and the Fire Protection Association of U.K. as well as the "Technical Guidelines for Fire Safety in Temporary Buildings in Construction Sites" issued by the SCDF.
- (b) The Contractor and his sub-contractors shall ensure not to construct specific deity altars and joss or paper burning practices at all times within the Project premises.
- (c) The Contractor shall ensure that all procedures, precautionary measures and safety standards stipulated in the Fire Safety Plan are implemented, communicated and complied with by all workers including sub-contractors and Interfacing Contractors.
- (d) The Contractor shall review and ensure the adequacy of the Fire Safety Plan as the Works progress.
- (e) The Contractor shall carry out monthly checks of firefighting equipment and test all alarms and detection devices installed on site. Tags/stickers shall be provided to indicate the monthly checks.
- (f) The Contractor shall conduct weekly inspections of escape routes, fire brigade access, firefighting facilities and work areas to ensure that the requirements stipulated in the Fire Safety Plan are complied with.

G2.23.36 Work in Confined Spaces

- (a) In addition to the requirements of the Workplace Safety and Health (Confined Spaces) Regulations and SS 568, Code of Practice for Confined Spaces, the Contractor shall also classify all manholes, enclosed formwork, culvert drains, excavations more than 4 m deep, partially enclosed excavations and tunnels as confined spaces and apply all legislative requirements of confined spaces.
- (b) The Contractor shall ensure that all persons who have to work in a manhole, tank or other confined spaces have attended the mandatory Safety Orientation Course (Manholes) approved by the Commissioner for Workplace Safety and Health.
- (c) The Contractor shall have controlled access/egress points to confined spaces to prevent unauthorised access. Where practicable the Contractor shall ensure that there are at least two readily accessible escape routes from each confined space.

- (d) The Contractor shall operate a tag system for entry so that all personnel entering the confined space can be accounted for.
- (e) The Contractor shall ensure that there is a certified man-riding cage capable of taking a stretcher and two persons, together with an identified crane equipped with rescue equipment, on standby at all times whilst work is carried out in the confined space. Where this is not reasonably practicable a stretcher which is capable of being brought manually out of the confined space should be located at a convenient point.
- (f) Gas monitoring shall be conducted by a competent confined space assessor to certify that the confined space is safe for workers to enter and thereafter at every four (4) hours intervals.
- (g) In addition, the Contractor shall ensure that suitable atmospheric monitoring devices such as anemometer and wet/dry bulb thermometer are made available for the competent confined space assessor to determine the air flow, ambient temperature and humidity level within the confined space.
- (h) The Contractor shall have controlled RFID tag access/egress points to confined spaces, shafts and tunnels to prevent unauthorized access. Where practicable, the Contractor shall ensure that there are at least two readily accessible escape routes from each confined space. Confine Space Attendant(s) (CSA) shall be present at access points for monitoring purposes at all times. The CSA shall hold a competent certification and trainings to perform his duties.
- (i) Where and when necessary the S.O. may require the Contractor to provide members of the S.O. and the Board with personal confined space monitoring devices. Maintenance, calibration and issuance shall be managed by the Contractor.

G2.23.37 Illumination

- (a) The Contractor shall provide temporary general illumination with a lighting level of not less than 100 lux unless otherwise stated for all work areas and comply with relevant WSH standards such as SS 531 Code of practice.
- (b) Site layout design plan shall incorporate points of illumination/spot lights to be placed to ensure sufficient coverage for low light work times. The S.O. may require the Contractor to install additional lighting.

G2.23.38 Welfare Provisions

- (a) Sanitary and washing facilities shall be provided in accordance with Code of Practice on Environmental Health and the Workplace Safety & Health Act. Toilet facilities shall be connected to a sewer/temporary septic tank with the approval of the Sewerage Department.
- (b) The Contractor shall provide suitable and sufficient temporary facilities on Site which are readily accessible taking into account the number and distribution of workers throughout all work locations.
- (c) These facilities shall include:
 - (i) Toilets and hand wash with hand soap areas;
 - (ii) A supply of clean drinking water; and

- (iii) Sheltered rest areas, to include seating, segregated from the worksite so that workers may safely remove helmets and other items of PPE. Such rest areas shall have sufficient waste bins.
- (d) These facilities shall be kept in a clean and serviceable condition and be available for use during all working hours. The locations may need to be air-conditioned if situated at bored tunnel and/or well ventilated if at station area.
- (e) In addition to constructed toilets, Portable toilets shall be provided where sewer connection mains are not available and regularly maintained at suitable locations where such can be considered as remote.
- (f) The Contractor shall prohibit handphone use in all work areas except in any designated areas set up by the Contractor sufficiently segregated from work areas and approved by S.O..
- (g) The Contractor shall prohibit smoking in all work areas except in any designated areas set up by the Contractor sufficiently segregated from work areas and approved by S.O.. Such areas shall be sheltered and have sufficient fire prevention and fighting facilities.

G2.23.39 Medical Facilities

- (a) The Contractor is responsible for the establishment of medical facilities at the site in compliance to local or international standards (Managing health for field operations by OGP/IPIECA ver 1/October 2011). A Medical Emergency Response Plan shall be established when planning the necessary establishment after review and approval by the S.O..
- (b) The Medical Emergency Response Plan shall integrate and cover medical treatment for emergency first aid assessments, treatment, stabilization, local medical evacuation and other necessary medical provisions to conserve the personnel life to the best of ability till he/she has reached nearest local hospital.

G2.23.40 First Aid

- (a) The Contractor shall provide sufficient numbers of qualified first aid for every 25 people for its works and size of its total workforce including sub-contractors. Such persons shall be responsible for responding to each incident involving injury or possible injury to any of Contractor's workers or staff. Such persons shall be on site at all times while Contractor is working. The Contractor's safety procedures shall define the role of such persons and responsibilities for contacting emergency services.
- (b) In carrying out any works requiring use of compressed air, the Contractor shall comply with the latest revision of the Workplace Safety and Health Act and its regulations, Workplace Safety and Health Guidelines for Compressed Air Works and other statutory requirements.
- (c) The Contractor shall also comply with BS 6164, Construction Industry Research and Information Association (CIRIA) Guidelines. For compressed air works, the Contractor shall provide medical locks, compressed air facilities and other equipment at the surface; along with suitably qualified compressed air works gauge attendant(s) and medical-lock attendants.

- (d) The Contractor shall engage their own medical officer (Factory Doctor) responsible for all medical aspects of the compressed air works including medical screening, keeping of records, compressed air medical treatment and any other medical related service. The medical officer shall be on standby when compressed air work is in progress.
- (e) It shall be the Contractor's responsibility to ensure that all personnel operating plant and equipment for compressed air works are suitably qualified and medically fit to work in compressed air.
- (f) The Contractor shall submit to the S.O. and to the Ministry of Manpower (MOM) the details of their compressed air works including plant system, method statement, decompression tables, medical and health records of workers working in compressed air, compression/decompression records and any other related records or items as required by the regulations.
- (g) An approved first aid station shall be provided and maintained at all times. The station shall be fully equipped to treat illness and injuries which can normally be expected to occur in work of the types required by this Contract. Medical supplies shall be stocked in the types and quantities recommended by the designated doctor.
- (h) The first aid station shall be located near the main access to the Site, readily accessible to ambulance service.
- (i) In addition, a number of trained first aiders and satellite first aid boxes or cupboards shall be provided and maintained to give effective first aid cover to the whole worksite, including any related off-site activities.
- (j) Where work is carried out during extended hours or on a shift system the Contractor shall ensure that there are sufficient trained first aiders on Site to give effective cover at all hours.
- (k) The first aid station and the satellite first aid boxes or cupboards shall be placed under the charge of HSE personals whom shall be trained in first aid treatment, and he, or a nominated qualified deputy, shall always be readily available during the hours where work is carried out on Site.
- (l) Contractor shall provide and maintain automated external defibrillators (AED). AEDs shall be readily accessible and all first aider are to be trained in its correct use.
- (m) Qualified first aiders shall be suitably identified with a logo of a green cross on their safety helmets.

G2.23.41 Ladders, Staircases and Man Lifts

- (a) The Contractor shall use step platforms instead of portable ladders for works at height subject to S.O.'s approval and establish a Permit-To-Work system for such works. In addition, for works more than 3 m, the Contractor shall demonstrate the stability of these step platforms to prevent toppling.
- (b) Ladders (step ladders and vertical ladders) shall comply to EN131 and local WSH guidelines. They shall be used for access only and shall not exceed 3 m in height.
- (c) The Contractor shall implement a step platform/ladder inspection procedure requiring an identification method displaying company name, unique number, inspection frequency and inspection status.

- (d) The Contractor shall provide at least one (1) steel staircase and one (1) lift as a minimum in each shaft when works have reached the depths of 4 m and then 10 m respectively to the approval of the S.O.. The steel staircase and lift shall be positioned in such a manner as to facilitate the evacuation of all personnel within ten (10) minutes via the staircase and lift from the work area. Scaffold staircases can be allowed where shaft depths have not reached 4 m. Upon approval of the S.O., the Contractor may use the permanent stairways in the shafts in place of the steel staircase(s).
- (e) At least 2 sets of emergency alarm shall be provided from independent power source, such that the second emergency alarm can be activated upon the first alarm failure.
- (f) The Contractor shall provide at least one (1) man-cage at each excavation shaft area to accommodate one (1) stretcher as a minimum for emergency rescue operation. The man-cage(s) shall be located at a place accessible or available within five (5) minutes from any accident location within the excavation area. The man-cage shall not be used to transport workers. Within the tunnel, a dedicated man-rider shall be provided to transport injured personnel.

G2.23.42 Scaffolds

- (a) The Contractor shall comply with the WSH Scaffold legislation. The Contractor shall employ an approved scaffold contractor registered with the Ministry of Manpower for the erection and dismantling of all scaffolding.
- (b) The Contractor shall develop a scaffold tagging system acceptable to the S.O. to indicate:
 - (i) Scaffolds under construction or demolition;
 - (ii) Scaffolds that are complete but have hazards associated with them; and
 - (iii) Scaffolds erected and safe for use.
- (c) The Contractor shall envelope all scaffolds with screen nets to prevent debris from falling outside the scaffold. Scaffold pipe ends and joint couplings to be capped off to prevent injuries. Nets shall be secured by nylon cable ties only.
- (d) Scaffold Regulations shall be also applicable to the erecting and dismantling of falseworks, temporary access ways using scaffold components and general barricades. Height of guard railings and barricades using scaffold components to be at least 1.3 meters high.
- (e) The Contractor shall use only proprietary access ladders and working platforms for system formworks. No mix and match using conventional catwalks and monkey ladders is allowed.
- (f) The scaffold contractor who is erecting, dismantling, or working on scaffolds must be approved by MOM as scaffold contractor. The scaffold contractor's workers must also receive training from MOM approved trainer. The workers shall not use scaffold systems for which they have not been trained and approved by MOM.
- (g) The scaffold supervisor approved by Ministry of Manpower (MOM) shall be assigned to direct and oversee the erection, dismantling, and use of scaffolds. Additionally, they must inspect scaffolds each day prior to use.

- (h) Stationary scaffolds over 30 m in height and rolling scaffolds over 18 m in height must be designed by a Professional Engineer (PE).
- (i) Scaffolds shall be designed by a PE and shall be constructed and loaded in accordance with that design.
- (j) A tag and permit system shall be used to inform personnel of the construction status of the scaffold. At a minimum, the system used shall inform users when a scaffold is complete and safe to be used and when a scaffold is under construction and is not ready to be used. When additional precautions are required to use the scaffold safely, for example, the use of fall protection systems, the system shall identify the precautions to be taken. The tag or permit shall be placed at each means of access to the scaffold. The competent shall be responsible for the tag and permit system.
- (k) A daily safety briefing shall be conducted with all scaffold personnel to discuss the work planned for the day and the health and safety requirements to be followed.
- (l) Scaffolds and scaffold components must be capable of supporting, without failure, their own weight and at least four (4) times their maximum intended load.
- (m) The site must be inspected to determine ground conditions, strength of supporting structure, and for proximity of electric power lines, overhead obstructions, wind conditions, the need for overhead protection or weather protection coverings.
- (n) Supported scaffolds must be set on base plates, mudsills, or other adequate firm foundation.
- (o) Frame spacing and mudsill size can only be determined after the total loads to be imposed on the scaffold and the strength of the supporting soil or structure are calculated and considered. This analysis must be done by a qualified person.
- (p) Base plates or screw jacks with base plates must be in firm contact with both the sills and the legs of the scaffolding. Compensate for uneven ground with screw jacks with base plates. Do not use unstable objects such as blocks, loose bricks, etc.
- (q) Scaffolds and scaffold components must be inspected for visible defects before each shift by a Scaffold Supervisor, and after each occurrence that could affect a scaffold's integrity (such as being struck by a crane).
- (r) Maintain scaffolding and materials (e.g., paint roller extensions, building material) in accordance with the minimum safe clearance distance from the overhead power lines as determined and approved by a licensed electrical engineer. All portable electric equipment must be protected by ground-fault circuit interrupters (ELCBs) or an assured equipment grounding conductor program.

G2.23.43 Suspended Scaffolding

- (a) Suspension scaffolds consist of one or more platforms suspended by ropes or other non-rigid means from an overhead structure(s). The common requirements for suspended scaffolds are addressed here; the competent person shall ensure scaffold type specific requirements are included as applicable.
- (b) The scaffold contractor who is erecting, dismantling, or working on scaffolds must be approved by MOM as scaffold contractor. The scaffold contractor's workers must also receive training from MOM approved trainer. The workers shall not use scaffold systems for which they have not been trained and approved by MOM.

- (c) The scaffold supervisor approved by MOM shall be assigned to direct and oversee the erection, dismantling, and use of scaffolds. Additionally, they must inspect scaffolds each day prior to use.
- (d) Scaffolds shall be designed by a PE and shall be constructed and loaded in accordance with that design.
- (e) A tag and permit system shall be used to inform personnel of the construction status of the scaffold. At a minimum, the system used shall inform users when a scaffold is complete and safe to be used and when a scaffold is under construction and is not ready to be used. When additional precautions are required to use the scaffold safely, for example, the use of fall protection systems, the system shall identify the precautions to be taken. The tag or permit shall be placed at each means of access to the scaffold. The competent shall be responsible for the tag and permit system.
- (f) A daily safety briefing shall be conducted with all scaffold personnel to discuss the work planned for the day and the health and safety requirements to be followed.
- (g) Scaffolds and scaffold components must be capable of supporting, without failure, their own weight and at least four (4) times their maximum intended load.
- (h) The site must be inspected to determine the strength of supporting structure, and for proximity of electric power lines, overhead obstructions, and wind conditions, the need for overhead protection or weather protection coverings.
- (i) Scaffolds and scaffold components must be inspected for visible defects before each shift by a competent person, and after each occurrence that could affect a scaffold's integrity (such as being struck by a crane).
- (j) Maintain scaffolding and materials (e.g., paint roller extensions, building material) at the minimum safe clearance distance from the overhead power lines as determined and approved by a licensed electrical worker.
- (k) All portable electric equipment must be protected by ground-fault circuit interrupters (ELCBs) or an assured equipment grounding conductor program.

G2.23.44 Fall Protection on Suspended Scaffolds

- (a) Each employee on a multi-point or two-point adjustable suspension scaffold must be protected by both a guardrail system and a personal fall arrest system.
- (b) Personal fall-arrest systems used on scaffolds shall be attached by lanyard to a vertical lifeline, horizontal lifeline, or scaffold structural member.
- (c) Guardrail systems must be installed along all open sides and ends of platforms and must be in place before the scaffold is released for use by employees other than erection/dismantling crews.

G2.23.45 General Safety

- (a) The Contractor shall provide, erect, maintain and finally remove, when instructed, Danger, Warning, Caution or Information signs, located appropriate to the site layout. The signs shall be no less than 1.5m x 1.0m in size written boldly in the four official languages. These shall be erected on existing footpaths and at points of access likely to be used by the public to warn or inform them of the existence of the Works. These notices shall be in addition to any notices required to be put up to meet statutory requirements.

- (b) The Contractor shall implement measures to eliminate dropped objects or falling objects such as identifying potential drop zones where there are works overhead with clear signages and barricades placed to divert personals. Tool bag and tool tether anchored securely should be implemented to prevent tools falling over. These implementations shall be additional to scaffold toe-board installation.
- (c) The Contractor shall ensure that all roads, pavements and public footpaths are kept clear of dust, silt and debris.
- (d) Unless otherwise agreed, the Contractor shall be responsible for the proper fencing, hoarding, lighting, guarding and watching of the Works. The Contractor shall also provide proper temporary roadways, footways, guards, fencing and hoardings so far as the same may be necessary for the accommodation and protection of the owners and occupiers of the adjacent property, the public and others.
- (e) All platforms, covers, ladders, stairways, staging, scaffolding and other provisions for access erected by the Contractor shall be installed in compliance with current legal requirements and made available for use as early as possible during the construction period. In cases where this is impracticable the Contractor shall provide all necessary temporary access facilities which shall be constructed, installed and maintained in a safe and secure manner.
- (f) Designated walkways along steel walers and struts shall be levelled, flush without tripping hazards and with rigid guardrails and toe boards securely provided.
- (g) Fall arrest system, including but not limited to life lines and anchors, shall be designed by a Professional Engineer (PE) and comply with the requirements of SS 528 and SS 570.
- (h) The Contractor shall provide capping on all protruding starter reinforcement bars with individual plastic/ rubber caps or with hose/tube.
- (i) The Contractor shall ensure that horseplay, practical jokes, scuffling, wrestling or fighting are strictly prohibited at the Site.
- (j) The Contractor shall ensure that the sale, keeping or consumption of liquor and prohibited substances on Site is prohibited.
- (k) No gambling, prostitution or other illegal or immoral activities shall be allowed anywhere else on Site.

G2.23.46 Safety (Anti-Fall) Net and Soft-Landing System

- (a) The Contractor shall provide and maintain safety net or soft-landing system in compliance to SS 292 or BS PAS 59 to catch persons falling whilst working in any location from where the risk of a fall cannot be eliminated. The system shall be of sufficient size and strength to catch any person for whose protection it is to be used and shall be so located to cover the area of the possible fall.
- (b) The Contractor shall conduct a sample test on the safety net or soft-landing system, comprising the net and its supporting structures, before it is installed. Subsequent tests shall be carried out when directed by the S.O..

G2.23.47 Barricade / Cover to Voids, Trench, Bored Holes and Open Edges of Structure

- (a) The Contractor shall barricade all excavation, bored holes, voids and open edges of structure under construction where a workman is liable to fall with secured and effective guardrails, barricades, and/coverings.
- (b) The Contractor shall provide “Danger” warning signs for barriers and barricades erected. All floor opening covers shall be stencilled or painted with “Danger, Do Not Remove”.
- (c) Where traffic flow is to be maintained over temporary road opening or crossing, the Contractor shall provide suitably designed chequered steel cover or decking over it. The Contractor’s Professional Engineer shall design and submit the cover or decking proposal to the S.O. for acceptance prior to commencement of the excavation.
- (d) The Contractor shall note that the slab openings on site have varying sizes from very small to very large and they are categorised into three (3) groups:
 - (i) Group 1: 300mm wide x any length
 - 1) Too narrow for a person to fall through, but wider than their foot and therefore possible to fall into with potential injuries to shin, knee, thigh or hip.
 - (ii) Group 2: 300mm – 1000mm wide x any length
 - 1) Big enough to fall through, with potentially serious consequences, but too small to be of use as a service opening. Sumps, pits and future access hatches are excluded from this group and shall be treated as Group 3 openings.
 - (iii) Group 3: Greater than 1000mm x any length
 - 1) Stairwells, air shafts and service openings are the most common in this category.
- (e) For each of the groups of openings, the Contractor shall comply with the ‘standard’ solution stipulated in this Section:
 - (i) Group 1: 300mm wide x any length
 - 1) A plywood cover (12 or 18mm thick) bolted or nailed down. Cover to be painted in a distinctive colour or pattern.
 - (ii) Group 2: 300mm – 1000mm wide x any length
 - 1) A13 mesh, fixed to the top steel and cast into the slab. After casting, a plywood cover can be fitted and secured to prevent debris from falling through;
 - 2) Mesh cast with slab provides immediate protection against fall and cannot be accidentally removed;
 - 3) Design load on A13 mesh has to be taken as 1.5KN. This is a commonly used load in ‘Dead and Live Load Tables’ for human traffic;
 - 4) Mesh is capable of spanning 1200mm under 1.5KN, but has been limited to 1000mm;

- 5) Mesh to be hot dipped galvanised;
 - 6) Embedment length to be 425mm in short span direction;
 - 7) Plywood Cover; and
 - 8) Cover fitted to prevent debris and materials from falling through. If cover is removed or becomes damaged the opening is still secure. Cover should be cut to fit inside the opening thus reducing the trip hazard and allowing other works to pass over it.
- (iii) Group 3: Greater than 1000mm x any length
- 1) Standard method of railings, toe boards and netting to be provided. Height of railings to be at least 1.3 m high;
 - 2) The Contractor shall note that the Stop-end below the mesh should be removed during striking of formwork. Stop-end above the mesh can be removed at the same time or left to serve as a stop-end for future screening works (if any). Plywood cover can be cut to suit;
 - 3) Removal of mesh shall be undertaken by the Contractor only when the opening is required by the relevant owner. As the relevant owner is usually maintain it is intended that the mesh be removed only when requested by them. It should not be removed during room handover;
 - 4) After removing the mesh, the Contractor should grind the remnant steel flush to the opening and make good to the exposed ends using an acceptable anti- corrosion treatment (e.g. cold galvanised paint, epoxy paint, etc);
 - 5) Sliding gates for protection shall be installed where appropriate. They shall be fully fabricated in factory prior to delivery and installation on site;
 - 6) In the process of installation, the contractor shall provide temporary restraint to prevent any movement (that can result in overrun and toppling) of the sliding gate until the installation with all safety features (E.g. stoppers to prevent over- run, gate safety sensors, etc) are completed;
 - 7) The work zone around the installation of the sliding gate must be cordoned off to prevent any unauthorized personnel from approaching;
 - 8) The manufacturer, supplier or builder of sliding gate must provide the operational manual for safe operation and maintenance of the gate;
 - 9) Gates should be checked and maintained periodically according to the manufacturer's or sub-contractor's recommendation. In addition, the track for the gate should be properly upkept to prevent any accumulation of debris which could cause the derailment of the gates; and
 - 10) The Contractor shall comply with recommendations from "Circular on Safety Considerations in Gate Design and Operation" issued by Ministry of Manpower and Building Construction Board dated 1st March 2012.

G2.23.48 Housekeeping

- (a) The Contractor shall implement a housekeeping plan approved by the S.O.. The plan shall propose the scope and coverage for housekeeping including but not limited to vector control, site cleanliness, organisation and storage of plant and materials, collection of waste, office decluttering, and recycling efforts.
- (b) The Contractor shall conduct weekly project wide housekeeping stand down schedule for two (2) hours on every Saturday morning or next work day if it is a public holiday, for housekeeping works. Senior management's active participation shall be visible. The S.O. may waive this requirement if the Contractor demonstrates high standards of housekeeping during the week.
- (c) This mandated weekly housekeeping schedule shall not exempt the Contractor from performing housekeeping works on a daily basis.
- (d) All materials shall be stored or stacked in a safe and orderly manner so as not to obstruct any passageway or place of work. Material piles shall be stored or stacked in such a manner as to ensure stability.

G2.23.49 Traffic Control and Road Safety

- (a) The C1A contractor shall provide and maintain all site access roads outside of the Contractor's site office/laydown area and work sites. The Contractor is to travel only on such designated site roads, which may change from time to time to suit the needs of the TWRP. The Contractor shall comply with all speed limits and traffic safety measures as issued by the S.O.. The S.O. from time to time may conduct traffic surveillance to audit compliance with speed limits, seat belt use and other traffic safety measures and may establish a system of fines or penalties for violations which shall apply to all Contractors and the workforce. The Contractor shall cooperate with and support such surveillance and compliance measures.
- (b) The S.O. shall impose speed monitoring devices such as Radar speed sign at strategic roads and other control measures to mitigate speeding vehicles.
- (c) The Contractor shall provide, install and maintain all necessary traffic and directional signs, barriers, blinkers, rotating beacons, cones, lane markings and inform the neighbours of road works or any road lane closure within its work areas.
- (d) The Contractor shall regularly maintain its site road surfaces to keep them free of potholes, unevenness, etc. Mill and patch method shall be required to repair any uneven surface defects.
- (e) All site vehicles, including 4 x 4 pick-up trucks / double cabs shall have use seatbelt policy labels prominently displayed at its front dash and rear passengers to comply.
- (f) The Contractor shall ensure only designated drivers operate the 4 x 4 pick-up trucks / double cabs within TWRP premises with a current photograph prominently displayed for inspection purposes. These vehicles shall be visually inspected and labelled Quarterly by a designated person. Vehicles shall be equipped with an adequately stocked first aid kit and serviced fire extinguisher.

- (g) All vehicles driven on Site shall be maintained in roadworthy condition and be registered with the appropriate authority in accordance with the Road Traffic Act. Each driver of these vehicles shall hold a valid driving licence authorising him to drive that class of vehicle. Such vehicles include but are not limited to cement lorries; pick-up and flat-bed trucks; lorry loader with articulating arms; and rough terrain forklifts.
- (h) Transportation of personnel on flat-bed trucks, cranes, forklifts, dumpers and similar vehicles not designed to carry passengers is prohibited.
- (i) Personnel transported on Site by pick-up trucks, lorries and similar shall be seated in the cab only. Free-standing chairs shall not be used as seating. All passengers must be seated at all times that the vehicle is in motion and shall not have any part of their body outside the vehicle. No more persons shall be carried in the cab of any vehicle than the number for which it is licensed.
- (j) The Contractor shall comply to the speed limit within the Site.
- (k) The Contractor shall appoint an appropriately trained spotter or banksman as traffic controller to control reversing vehicles and congested machinery movement. The traffic controller shall wear a high visibility two-piece luminous yellow uniform clearly marked "Traffic Controller" or similar. At low visibility situations, the Contractor shall provide to his Traffic Controller blinking LED vests and batons.
- (l) The Contractor shall ensure that the drivers of all delivery vehicles to and from Site wear appropriate PPE including, but not limited to, safety footwear and safety helmets outside the cabs of their vehicles.
- (m) All roads within Contractor site or work areas used by mobile plant and vehicles shall be constructed of hard standing and suitable for its intended purpose.
- (n) No unauthorised parking shall be permitted anywhere other than at the designated parking area. Only cars bearing the permit issued by the S.O. and authorised visitors shall be allowed to park at the designated parking area.
- (o) Consideration shall be given at all times to the safety of the road users and gates should be positioned so that they minimise the additional risk to traffic at such locations as road junctions, bends, etc.
- (p) A clear line of sight shall be maintained for all drivers of vehicles using the Contractor's roads, taking into consideration the speed limit of that road and the position of existing street furniture and trees / vegetation, to maximise the warning distance of approach.
- (q) If deemed necessary by the S.O. suitable mirrors shall be positioned at these points to enhance vision of traffic movement both on the roadway and on entering / leaving the Site.
- (r) If deemed necessary by the S.O., a traffic lane may be coned off to allow for safe deceleration.
- (s) Warning signs in compliance with the Code of Practice for Traffic Control at Work Zone shall be conspicuously displayed at appropriate distances before such gates into the Site as to give all drivers a clear understanding of the traffic hazard ahead.

- (t) If vehicles entering / leaving the Site have to cross a public footpath or pavement then a worker must be deployed to control pedestrians as well as road traffic. Suitable warning signs should be deployed to alert pedestrians to possible traffic movement across the footpath.

G2.23.50 Tunnel Safety

- (a) The Contractor shall develop and submit a Tunnel Safety Management Plan where it is relevant to his scope of works and to submit to the S.O. for approval. The following shall serve as a guide while developing the Tunnel Safety Management Plan:
 - (i) Scope of tunnelling works;
 - (ii) Identify hazards working in a tunnel;
 - (iii) Safety procedures when operating tunnelling equipment and machinery;
 - (iv) Procedures of work safely in confined spaces;
 - (v) Safety procedure when doing hot work and working in compressed air;
 - (vi) Illumination needs in the event of power supply failure;
 - (vii) Emergency and precautionary measures associated with tunnelling works;
 - (viii) Entry and exit point controls and accountability of persons in the tunnel;
 - (ix) Provision of telephone network or means of communication; and
 - (x) Medical check-up for working in compressed chamber.
- (b) Where applicable the Contractor shall appoint full time one (1) WSHS for every tunnel boring or pipe jacking machine deployed in his project.

G2.23.51 HSE Cross Border Sharing

- (a) The Contractor shall facilitate and host visits by other contractors to his worksites and to share his HSE management experiences with them.
- (b) The Contractor's management team shall attend visits to other contractor worksites with good HSE management initiatives and skills when directed by the S.O. and to learn and implement the good initiatives at his site.

G2.23.52 Use of Explosive Material for Purpose of Blasting, Demolition or Removal of Rocks

- (a) Blasting proposal and operations must be carried out by a qualified company and personnel which shall have valid Licenses / Certifications compliant with legal requirements in Singapore, such as:
 - (i) License to possess or control explosives;
 - (ii) License to store explosives;
 - (iii) License to conduct blasting operations; and

(iv) License to deal with Explosives

The area where explosives are to be used shall be clearly defined with warning notices and barriers located to prevent unauthorised entry and to protect the outside areas from flying debris. There shall be a procedure for informing neighbours if there is any likelihood of an explosion causing alarm. During detonation of explosives look-outs shall be posted in safe position in radio or sight communication with the shot fired. Demolition by means of explosives shall be outside normal work hours. Blasting proposal submissions shall be submitted to S.O. for approval prior project deployment.

- (b) The Contractor shall provide provisions for blast zone warning signal system to ensure unauthorized entry is avoided. Re-direct vehicle access or personal access where necessary. To maintain intercommunication between blast zone guards and blast firer to ensure area is clear prior to firing. Only approved radio transmitters are allowed within the blast area.
- (c) The Contractor shall exercise the utmost care not to endanger life and property by making proper use of blasting mats and other protective devices. Should property damage occur, make restoration as required at no additional cost to borne by other party.
- (d) The Blasting crew shall consist of blaster in-charge, blast setters and fire watch. The blaster in-charge shall hold pre-blast meeting with relevant stakeholders to brief the proposed and preparation blast plan.
- (e) The Contractor shall submit a close-out submittal and blast log, on completion of blasting works at end of every day. Is to ensure all blasting related equipment and materials to be removed and detonated or unloaded explosives had been properly disposed of.
- (f) Stores shall be clearly marked with appropriate safety signs, locked and at all times kept dry, well kept, free from all other chemicals and well away from any flammable hazardous areas.

G2.23.53 Lifting Operations

(a) General

- (i) The Contractor shall be responsible for ensuring that all lifting operations carried out on site and any rental cranes brought onto site are done so with minimal risk of injury to persons including members of the public or damage to property.
- (ii) Effective control must be exercised at all stages of a lifting operation through the deployment of a competent lifting engineer, lifting supervisor, a qualified experienced signaller, a registered crane operator and qualified experienced rigger.

- (iii) All lifting machines, lifting appliances and lifting gears used on site must be suitable for the task performed and certified safe for use. The lifting machines, appliances and gears are to be used within their rated safe load capacity and must be in good repair.
- (iv) The Contractor is to submit a weekly list to the S.O. of all cranes deployed on site detailing their ownership, make, identification number, maximum safe working load and the corresponding radius, lifting machine (LM), lifting appliance and lifting gear certificate number and date of expiry.
- (v) The Contractor shall cooperate with the S.O. in implementing and maintaining a craneage coordination plan for the site. Such cooperation may include, but not be limited to, providing plans and forecasts for all the Contractor's crane locations, durations, lifts, operating times, swing radii and other information requested by the S.O. to ensure that the crane operations of all contractors are coordinated to avoid conflicts. When necessary, the Contractor shall revise its crane locations or operations to maintain such coordination and safe operations.
- (vi) The Contractor shall ensure that any lorry loader with articulating arm on site is only used for the delivery to Site and collection and removal from site of materials within its rated lifting capacity, and not for lowering any materials into a trench or excavation, or for raising any materials to a higher level. However, transporting materials or plant and machinery around Site may be permitted if the lorry loader with articulating arm has been tested on the Site by an Authorised Examiner.
- (vii) All lorry loaders with articulating arm entering into the Site shall have an interlocking system installed to prevent a lorry loader with articulating arm from moving off before its articulated boom is fully retracted back to its "stored" position.
- (viii) The boom of any lorry loader with articulating arm shall not be used for holding down materials or equipment and it shall be retracted to its "stored" position before it moves off.
- (ix) Loading and unloading operations involving lorry loader with articulating arm shall take place over the side of the main body and not in an arc over either the front or rear of the chassis, in accordance with safe working practices.
- (x) No excavator shall be used as a lifting machine on site unless it is originally designed and manufactured to also function as a lifting machine and comply with all MOM stipulated requirements.
- (xi) The Contractor shall ensure that a permit to lift system is operated to evaluate all routine planned lifting operations generically and all heavy or one-off lifting operations individually. This latter category of lifting operations includes the raising and lowering of items of plant in excavations. Lifting plans shall be submitted to the S.O. for acceptance in advance of any heavy or one-off lifting operations.

- (xii) Lifting operations with the use of cranes and excavators approved for lifting, shall be issued with a daily lifting pass by the Contractor after the lifting plans and permits-to-work have been verified to be in order by the Contractor's Lifting Engineer. Lifting operations can only be performed with a valid lifting pass. The Lifting Pass (A4 sized and in bright green) shall be visibly displayed in the operator's cabin. The Contractor shall craft and submit a Lifting Pass format for the S.O.'s approval prior to use. The S.O. will check the lifting permits issued on a daily basis and in the absence of a lifting plan, the lifting permit will be revoked and a stop work order will be enforced by the S.O. until the unsafe condition is corrected.
 - (xiii) The Contractor shall have suitable medical devices to perform daily health screening on all of its crane operators and have a crane operations log book or equivalent to record on daily basis. The records shall be kept at Contractor's Site office. The Board and the S.O. may request for such information.
 - (xiv) Lifting supervisors, lifting engineers and riggers shall wear brown, white and red safety helmets respectively. Riggers and signalmen shall wear red reflective vest/uniform and designation tags.
 - (xv) The crane operator is to ensure that the outriggers are fully extended, and any adjustments that are made to the jacks to level the crane before the lifting operation commences. All lifting machine with outriggers, including lorry loader with articulating arm, shall have steel plates of minimum dimension 1m by 1m by 25mm placed under all the outriggers deployed for a lifting operation unless that crane is entirely sited on hard standing such as a reinforced concrete surface, with no void underneath. Pieces of timber are not to be used.
 - (xvi) Lifting of machineries and equipment shall be deployed as per manufacturer's recommendations.
 - (xvii) Lifting plans shall identify centre of gravity and lifting points for each lift.
- (b) Lifting Machinery
- (i) This term includes, but is not limited to, lattice jibbed crawler mounted cranes, hydraulic variable jib mobile cranes, rail mounted gantry cranes, mobile and fixed tower cranes.
 - (ii) The Contractor shall ensure that before any lifting machine including lorry loaders with articulating arm used for delivery to site, is brought for use on Site; the labels on all controls, the details on the safe working load radius chart and any other safety related notices in the cab or on the body of the machine shall be written in English as well as in a language comprehensible to the crane operator.
 - (iii) Before being taken into use on site for the first time, in the case of a lifting machine which undergoes assembly on site, or following the substantial alteration of any type of lifting machine such as the jacking up of a tower crane, the lifting machine must be tested by an authorized examiner and subsequently every six (6) months whether it is owned by the Contractor, one of his subcontractors or by a crane rental company. The Contractor shall arrange for the testing before the expiry date so that there is continuity of coverage of LM certificate.

- (iv) The Contractor shall ensure that no lifting operation shall be carried out on site using the auxiliary hook of a mobile crane unless the SWL of this is shown on the LM Certificate in addition to that of the main hook block and is not exceeded.
- (v) The Contractor shall ensure that both the lifting supervisor and the crane operator are able to understand the working load chart.
- (vi) The age of lifting machinery providing mechanical lifting works shall be in accordance with MOM's and relevant authorities' requirements.
- (vii) Cranes fitted with a load radius indicator (LRI) shall sound an audible alarm in the crane cab if it's safe working load is exceeded on either the main or the auxiliary hook. A second alarm connected to the LRI, shall be fitted external to the cab and shall emit a signal of a sufficient volume to make it audible above the ambient site noise levels during working hours. Visual warning shall also be provided externally to indicate safe working range and overload conditions.
- (viii) LRI shall be fitted with a limiting device, which disables the crane from continuing with any lifting operation under overload conditions. Once disabled, the device should only permit the crane to return to the safe working range. The device shall be tamper proof, with no over-ride to disable it, and shall be maintained in good working order. In case of provision of bypass switch for LRI, it shall be secured by a lock and kept with the lifting supervisor.
- (ix) LRI shall be calibrated 6-monthly by an approved agency and verified by the Authorised Examiner during his 6-monthly inspections. Records of LRI calibration shall be submitted to the S.O.. If there is any doubt as to the accuracy of the LRI the machine shall be taken out of use until the calibration is carried out to the satisfaction of the S.O..
- (x) If the crane is down rated by the Authorised Examiner during his 6 monthly inspections then the LRI shall also be calibrated and a new capacity chart should be drawn up and posted in the crane cab. The Examiner shall highlight the above details on the LM certificate.
- (xi) Lifting machine shall not be used for any unsafe operation that may affect its overall integrity or stability. The Contractor shall comply with Ministry of Manpower regulation and circulars regarding fitting of cranes with data loggers.
- (xii) The Contractor shall ensure that every lifting machine capable of travelling or tracking, including a gantry crane, or a slewing motion, such as a mobile crane, shall be operated in such a manner that there is always an unobstructed passageway in excess of 750mm between it and any fixture or other machinery.
- (xiii) All Lifting machines that operate on site shall be installed with a front/rear-view camera that enables the operator to have a clear view of the back of the machine and visual warning light (Green, Amber, Red) to indicate safe working load range and overload conditions.
- (xiv) The use of fly jib of any Lifting machines shall be subjected to the approval of the S.O..

- (xv) The concurrent use of the crane's main and auxiliary hooks during lifting shall be prohibited.

(c) Crane Data Logger and Stability Control System

- (i) All cranes without manufacturer fitted data loggers operating on site shall be retrofitted with data loggers approved by the S.O.;
- (ii) All cranes with manufacturer fitted data loggers shall have the data loggers enabled when operating on site;
- (iii) The Contractor shall provide remote real-time crane monitoring on all cranes expected to be in operation for more than one (1) month. Temporary rental cranes are exempted from requirement for remote real-time monitoring.
- (iv) The Contractor shall ensure that the data logger is calibrated by authorised personnel from the data logger manufacturer when the crane is mobilised on site for the first time and subsequently on a six-monthly basis. A calibration certificate shall be submitted to the S.O.;
- (v) The data logger shall have the following capabilities:
 - 1) Detection of events of overloading, over-hoisting, over-derricking and bypassing of limit switches;
 - 2) Continuous recording of critical crane operational parameters;
 - 3) Data downloading feature;
 - 4) Data security and anti-tampering;
 - 5) Real time stamping;
 - 6) Report generation feature;
 - 7) Graph generation feature;
 - 8) Warning and alert feature; and
 - 9) Remote real-time monitoring feature.
- (vi) The data logger shall be able to perform continuous record of the following crane operational parameters whenever the crane is in operation:
 - 1) Date and time of the operational hours of the mobile crane, from the time engine starts till engine shuts;
 - 2) All lifting operations which includes overloading, over-hoisting, over-derricking and bypassing of limit switches;
 - 3) Load Moment Indication;
 - 4) Weight of load lifted;
 - 5) Crane operating capacity (Safe Working Load);
 - 6) Slewing angle;
 - 7) Boom angle;
 - 8) Boom Length;
 - 9) Crane operating radius;

- 10) Number of part lines indicating hook configuration (Main or Auxiliary hook);
 - 11) Crane operational hours since installation;
 - 12) Faults/settings display; and
 - 13) Position of jib or boom.
- (vii) The data recorded by the data loggers shall be remotely monitored, downloaded and interpreted on a monthly basis and submitted to the S.O. in the form of a report. The S.O. may at his discretion require the Contractor to download the data when he deems necessary. The monthly data logger report shall include the following details together with the set of recorded operational parameters:
- 1) Project contract number;
 - 2) Name of crane operator;
 - 3) Manufacturer of crane;
 - 4) Model number of crane;
 - 5) Serial number of data logger;
 - 6) Type of crane (e.g. crawler or telescopic);
 - 7) LM certificate of crane;
 - 8) Crane operating hours
 - 9) Schedule of crane servicing
 - 10) LRI chart of crane;
 - 11) Summary of overloading, over-hoisting, over-derricking and bypassing of limit switches events; and
 - 12) Continuous graphical plot of load moment and boom radius data against time.
- (viii) Stability Control System Performance Outcome
- 1) All lorry cranes shall be equipped with Stability Control System (SCS) such that during the operation of the crane, the SCS will:
 - a) Continuously monitor the stability of the lorry crane, such as through detection of its outriggers position and the weight of the load it was carrying; and
 - b) Upon detection of any unsafe situation such as over-moment or instability of the lorry crane, immediately prohibits further operation of the crane, except to allow manoeuvring to bring the crane into the safe zone.
 - 2) The SCS shall be equipped with both visual and audio alarm to warn operator and persons in the vicinity of the crane when the SCS detects any unsafe situation.
 - 3) The SCS shall be tamper-proof, and not be easily bypassed.

- 4) The SCS shall be designed such that in the event of sensors or system failure, the crane shall not be able to operate until the fault has been rectified.

(d) Erection of Cranes on Site

- (i) The Contractor shall engage only approved crane contractors to install, alter, repair or dismantle any parts of a mobile or tower crane which affect the lifting capacity of that crane. A copy of the MOM approval letter shall be retained with the crane erector doing the repair.
- (ii) The minimum strength of the team who erects / dismantles any tower crane must be one approved crane erector and five trained assistants, and for a mobile crane is one approved crane erector and two assistants.

(e) Lifting Gear (LG) / Lifting Appliance (LA)

- (i) The Contractor shall ensure that every LG/LA brought onto site, including that accompanying rental cranes, lorry loader with articulating arm, excavators, cement buckets, air receivers, skips, welding sets etc. has a valid LG/LA certificate and clearly marked with its SWL. All LG/LA shall be inspected by an Authorised Examiner once every six (6) months.
- (ii) Chain sling type of Lifting gear shall not be used and instead shall be of the webbing/round sling (Nylon/Polyester) and wire rope type (Steel).
- (iii) Lifting shackles shall be the type where split cotter pin can be applied before making a lift. All lifting shackles shall be checked for deformity daily. Where the Lifting Gear is supplied by the manufacturer or supplier, to meet these requirements shall be ensured by the Contractor at his own cost prior to use.
- (iv) The Contractor shall ensure that LG/LA is not loaded beyond its SWL and this includes multi slings being used at variable angles.
- (v) The Contractor shall ensure that LG/LA is not used for any purpose other than the raising or lowering of a load. If an excluded activity takes place, such as the use of a lifting chain for towing an item of plant or machinery, then the Contractor shall ensure that this item of LG/LA is no longer used for any further lifting operations.
- (vi) If an item of lifting gear is inadvertently subjected to a force exceeding that it is designed to experience when lifting an object at its SWL then the Contractor shall ensure that it is taken out of use.
- (vii) The Contractor shall implement an inspection programme to thoroughly check all LG/LA by a LS prior to its first use on site and thereafter on a monthly basis. A monthly colour coding system shall be adopted. Defective LG/LA shall be discarded.
- (viii) When not in use the Contractor shall ensure that all items of LG/LA are stored in a rack sheltered from the weather and maintained regularly. Should any LG/LA be exposed to a corrosive material e.g. wet concrete, it must be washed off afterwards and re-greased.

(f) Lifting Supervisors (LS)

- (i) The Contractor shall ensure that a sufficient number of qualified LS are employed on site, whether by himself or by his subcontractors, to give adequate cover for all lifting operations carried out both by day and by night including loading/unloading. There must be one LS present for one crane including all minor and major operations. In addition, the contractor shall also station at least one additional worker to assist signaller for lifting operations carried out near public areas.
- (ii) The qualified LS must meet a minimum of three (3) years' experience performing LS duties and has attended MOM approved lifting supervisor course to oversee the activities of the various lifting operation on site. The LS will assist the lifting engineer to ensure approved lifting plans are in place and compliant. The Contractor shall ensure his Permit to Hoist/Lift shall have checked and endorsed by his LS.
- (iii) The Contractor shall install warning devices/ flags at least 2 m above top of hoarding that are alongside roads, footpaths and adjacent structure.
- (iv) LS shall supervise and co-ordinate all lifting operations under his/her charge. LS shall familiarise himself with the safe working load chart of each crane for which he is responsible and has identified himself to the operator of each such crane as being the only individual whose instructions concerning any lifting operation are to be followed. This includes the positioning of the crane prior to the lift as well as the slinging of the load.
- (v) Before any lifting operation involving a mobile or crawler mounted crane is carried out, the LS must satisfy himself that the crane is positioned suitably close to the load and its destination to ensure that the operation can be carried out at the safest appropriate radius.
- (vi) The LS shall ensure that the load is safely rigged, and a tag line is attached if appropriate, before signaling to the crane operator to start the lift. The LS is responsible for the load until it is safely resting at the intended destination either by taking control of the operation himself for non-routine lifts or, for routine lifting operations, by thoroughly briefing the crane operator, riggers and signaller on the safe procedure to be followed.
- (vii) The Contractor shall put in measures to discipline any person other than the dedicated LS, or one acting under his close supervision, who attempts to take control of any lifting operation other than those of a routine nature where a safe lifting procedure has already been established.

(g) Crane Maintenance Supervisor

- (i) The Contractor shall appoint a full time Crane Maintenance Supervisor certified competent by MOM approved crane supplier company to attend to all the mechanical aspects of operations involving lifting machines, lifting equipment and lifting gear on site.
- (ii) The Crane Maintenance Supervisor shall carry out periodic planned maintenance on all lifting machine to ensure that the lifting machine is in good working order whilst in the S.O. sites.

- (iii) The Crane Maintenance Supervisor shall be responsible for ensuring that each crane operator is competent to carry out the checks necessary before the crane is taken into use on any day / shift and shall carry out his own weekly thorough check of all cranes on site.
- (iv) The Crane Maintenance Supervisor shall ensure where there is a need for maintenance personals to conduct machinery repair works within the project, the maintenance personals shall have undergone adequate trainings by the Contractor or outsourced machinery repair company. Mechanic Training and risk assessment materials shall be provided to the S.O. for acceptance.
- (h) Crane Operator
 - (i) The Contractor shall engage only qualified crane operators with at least:
 - 1) Five (5) years of experience in operating similar types of cranes with no record of crane toppling / failure or barred from any site previously for a crane related incident;
 - 2) The crane operator shall enter the date, types of maintenance carried out and any malfunction of the crane in a checklist or logbook. He shall not operate the crane until any such defect is rectified, and the crane's use is authorised by his LS; and
 - 3) All cranes shall be checked by its operator at the start of any day / shift using a checklist written in English and in a language comprehensible to the operator. Copies of the checklist together with LM certificate, operator certificate, permits should be retained in the crane cabin for verification.
 - (i) Lifting Engineer (LE)
 - (i) To ensure that day to day lifting operations are monitored at an appropriate level of management, the Contractor shall appoint lifting engineer with a minimum of five (5) years site experience and has attended MOM or Institute of Engineers Singapore conducted Appointed Person – Lifting Operations course to oversee the activities of the various lifting operation on site. All consultations and problems concerning safety of any lifting operation shall be referred to this person for a final decision.
 - (ii) The Contractor shall ensure that a lifting plan is drawn up and agreed with his appointed LE responsible for lifting operations before any lifting operation is started. The LE shall verify that the lifting supervisor, the crane operator, the signalman and the riggers understand the part they have to play in ensuring that it is carried out safely. The lifting plan shall be forwarded to the S.O. for acceptance.
 - (iii) The Contractor shall ensure crane access routes and working areas are designed by a relevant Professional Engineer (PE). The PE shall provide layout plans that highlight any potential danger areas to avoid and indicate access routes and safe working areas for the types of mobile cranes to be used. The LE shall ensure the PE's requirements are always complied to.

- (iv) The LE is responsible for checking that the proposed location of the crane for the lifting operation is sufficiently compacted to bear safely the force exerted by the crane, taking into consideration the proposed load to be lifted, and any additional forces exerted by the crane slewing or derricking.
- (v) The Contractor shall put in place a system for inspecting and upgrading the route over which a crawler mounted / mobile crane is to transit when it is being moved on the site before any proposed lifting operation to ensure stability of the crane whilst travelling. Inspections must be conducted at least once a week or after inclement weather. The records of the inspections must be properly documented. This procedure shall also be followed when it is proposed that a crane transits carrying a load.
- (vi) The LE shall certify in writing, at least once a day that the crane access routes will support the force exerted by each crane and any load it may be carrying. The LE is to make an entry on a form designed for that purpose, endorse it and hand it to the operator to keep in his cab before any crane is transited. Crane access checks shall be repeated after inclement weather as appropriate.
- (j) Mechanical Lifting
 - (i) Mechanical crane operations, banking, slinging shall be performed by competent persons, who have been assessed, inducted and authorised by the Contractor. Lifting crew induction and behavioural assessment shall be devised and submitted to the S.O. for approval.
 - (ii) Besides routine lifting works, the Contractor shall assess potential heavy or critical lifts early in the planning of the construction activities, to decide if additional lifting plans and rigging studies are required if it is a non-routine heavy/critical lift. The following of any factor shall be classified as heavy/critical lifts:
 - 1) Load greater than 20 tonnes;
 - 2) Crane with capacity greater than 150 tonnes;
 - 3) Lift is greater than 80% of crane capacity;
 - 4) Tandem lift by two cranes where lift exceeds 50% of the capacity of either crane;
 - 5) Lifting over live process equipment or pipes or active work area; and
 - 6) Crane with a boom or jib greater than 100 meters.
 - (iii) The Contractor shall ensure all cranes classified under the critical lift factor, Tower cranes and Gantry cranes are compliant to BS, BS EN, ANSI/ASME or EU regulations. The Contractor shall submit the crane's manufacturing Declaration of Conformity certification when submitting the origin of manufacture of all cranes to the S.O for approval.
- (k) Temporary Rental Cranes
 - (i) Temporary Rental Cranes are types which will be deemed by the Contractor to operate on the project for less than one (1) month.

- (ii) The Contractor must submit to S.O. at least one (1) day before any rental mobile / crawler mounted crane is brought to the Site:
 - 1) The lifting plan for the operation to be carried out;
 - 2) A copy of the current LM certificate for the crane;
 - 3) A copy of the crane operator's MOM registration;
 - 4) The name of the LS will be responsible for it;
 - 5) A record of any repairs carried out since the last LM certificate was issued; and
 - 6) A copy of the most recent data logger report for the crane.
- (iii) The Contractor shall ensure that, as far as practicable, rental cranes coming onto Site are manned with the same operator, who shall undergo site induction training before his first job on Site. If for any reason the crane rental company sends a substitute operator then he also shall undergo site induction training.
- (iv) Before any crane is accepted onto Site for use it must undergo a thorough mechanical check by the Crane Maintenance Supervisor, and the findings verified by the Contractor's HSE personal.
- (l) In Cabin Cameras for Excavators and Lifting Machines
 - (i) All excavators and lifting machines shall be installed with in-cabin cameras;
 - (ii) In-cabin cameras shall be installed with forward and rear facing lens connecting to a monitor;
 - (iii) The cameras shall auto record when the excavator and/or lifting machines are powered up;
 - (iv) The camera system shall be connected and adapted (due to difference in voltages between camera and machine batteries) to the electrical panel directly;
 - (v) The camera shall come with a LCD or LED monitor and minimally a 64GB memory card. For SD cards, the specifications shall minimally be that of Class 10;
 - (vi) The camera lens shall have minimally full HD resolution of 1920x1080 pixels;
 - (vii) The cameras shall have the following recording functions:
 - 1) Auto recording function when engine is on; and
 - 2) Auto overwriting when recording memory is full.
 - (viii) Recordings shall include date, time, images and sound. GPS signals shall be used for auto-calibrating time set-up. If GPS is not available, the built-in clock shall be used;
 - (ix) The camera system shall have a function to allow the complete disabling of any built in G sensors;
 - (x) The camera shall have a view angle of at least 120 degrees;

- (xi) The camera shall be equipped with Auto and Manual dimmer features for night viewing as well as built in sun visor;
- (xii) As part of daily checklist before operating an excavator used for lifting purposes, the excavator operator shall ensure that the in-cabin camera and its memory card are in place and functioning properly. The excavator shall not be operated if the in-cabin camera or its memory card is faulty;
- (xiii) Only a person authorised by the Project Manager shall be allowed to remove the memory card from the memory card slot; and
- (xiv) To prevent data tampering and unauthorised removal of memory card, the memory card shall be sealed in the memory card slot of the in-cabin camera.

G2.23.54 Movement of Heavy Machinery

- (a) All heavy machineries brought into the Site are subjected to the Safe Work Procedures for Movement of Heavy Machineries.
- (b) This Safe Work Procedure requires the Contractor to implement three (3) control points prior to moving and using of the heavy machineries on sites:
 - (i) Permit to Move;
 - (ii) Permit to Work; and
 - (iii) Resuming Work after Heavy Rain.
- (c) Heavy Machinery refers to any equipment with a high Centre of Gravity (CG), including all Lifting Machineries (LM) such as Boring Rigs, Trench Cutters, Grouting Machines (e.g. Deep Soil Mixing Machines, Wet Soil Mixing Machines), rotary percussion rigs and D-Wall machines.
- (d) The Permit to Move is aimed to monitor and control the movement of large, high mast machines to ensure that they are on firm, well prepared ground at all times.
- (e) Prior to moving a Heavy Machinery, a Permit to Move is required. The subcontractor needs to prepare and submit a Permit to Move to the Contractor. The Permit shall be valid for one (1) rig and only for the location prepared and tested. Each new location shall require a separate permit.
- (f) Prior to the approval of the Permit to Move, the following requirements for the location where the heavy machineries are to be moved to shall be satisfied:
 - (i) Ground to well compacted using 10T Roller or equivalent and tested (Plate Bearing Test or calibrated Cone Penetrometer Test);
 - (ii) Any soft spots to be excavated, replaced with suitable soil, re-compacted and retested;
 - (iii) Ground then to be topped with compacted hardcore. Minimum 500mm or more if required by the PE;
 - (iv) Steel plates to be laid on top of hardcore;
 - (v) All testing to be witnessed by relevant PE;
 - (vi) PE to issue Certificate of Supervision (CoS) once works are completed;

- (vii) CoS should include a signed statement by the supervising PE, design calculations, plan showing area covered by CoS, copy of test results; and
- (viii) Area covered by CoS to be not greater than 300m². Relevant documentations need to be attached with the Permit to Move.
- (g) The Permit to Move shall be prepared, prepared by the subcontractor to the Contractor. It shall be approved by the Contractor (recommended 3 tiers of approval i.e. lifting engineer, authorised WSH Personnel and Project Manager/ Authorised Manager), acknowledged by QP(S) and audited by the S.O..
- (h) The PE Design Calculations must be comprehensive and should allow for the following as a minimum:
 - (i) Static plus dynamic loads;
 - (ii) Non-uniform distribution of load; and
 - (iii) Appropriate allowances for flexing of steel plates.
- (i) Permit to Work and daily checklists are required for daily operations prior to commencing work. The application of the permit shall be accompanied with relevant documentations (i.e. CoS by PE, machine daily checklist).
- (j) The Permit to Work shall be prepared by the subcontractor and submitted to the Contractor for approval. It shall be approved by the Contractor (recommended 3 tiers of approval i.e. Works Supervisor, WSH Assessor and Project Manager/ Authorised Manager). After periods of heavy rain, the following measures are required:
 - (i) Steel plates to be lifted and ground inspected for loss of integrity by Lifting Engineer;
 - (ii) Surrounding ground to be also inspected for localised flooding which could reduce the ground's bearing capacity; and
 - (iii) QP(S) to witness the inspection and approve prior to resuming the work. If the ground shows loss of integrity then CoS shall be void and preparation work must be repeated.
- (k) The Lifting Engineer must complete and sign a crane access checklist on a daily basis and after heavy downpour. Following are the examples of items to be checked in the crane access checklist:
 - (i) Laying of hardcore;
 - (ii) Compaction of Subgrade;
 - (iii) Laying of steel plates;
 - (iv) Dimension in accordance to layout and details;
 - (v) Proper drainage;
 - (vi) Stability;
 - (vii) Barrier next to ground depression or excavated area; and

- (viii) Other Precautions – Verticality (DSM & Boring rigs only). Vertical stability varies between machines and direction of tilt. Typically, it is around 70. It is recommended to limit the mast tilt during movement to 50 (i.e. 20 less than lowest limit of vertical stability).

G2.23.55 Worker and Site Transport

(a) General

- (i) All vehicles driven on Site shall be maintained in roadworthy condition and be registered with the appropriate authority in accordance with the Road Traffic Act. Each driver of these vehicles shall hold a valid Singapore driving license authorising him to drive that class of vehicle. Such vehicles include, but are not limited to cement lorries, pick-up and flat-bed trucks; lorry loader with articulating arms; and rough terrain forklifts.
- (ii) Transportation of personnel on flat-bed trucks, cranes, forklifts, dumpers and similar vehicles not designed to carry passengers is prohibited.
- (iii) Use of two/three-wheeler devices such as motorcycles, personal mobility devices and bicycles are prohibited to be used or parked within TVB. These devices shall not be used within the project and common areas and are to be parked out of TVB.
- (iv) The Contractor shall provide buses as the mode of transport for workers from dormitories or other locations to the Site and return. All workers being transported shall be seated at all times. The Contractor shall coordinate with their designated transport provider on timings. Parking and waiting by the roadside is not allowed. There will be no space outside of the Contractor's assigned site to park buses or other vehicles.
- (v) The Contractor will be assigned a limited number of permits allowing personal vehicles, not including motorcycles and bicycles (including motorized bicycles, scooters and similar), to drive to and from the Site. The number and duration of permits will be determined by the S.O. and may change during the Contract Period if determined necessary by the S.O. to maintain efficient flow of traffic to, from and within the TVB.
- (vi) The C1A contractor will provide and maintain TVB roads, outside of the Site and establish necessary road traffic rules and procedures, including setting an appropriate speed limit. The Contractor shall comply with all rules and regulations. The following provisions apply to those roads and transport within the Contractor's own site and work areas.
- (vii) Any routes on the site where headroom is restricted shall have appropriate warning signs posted at the approach to such restriction.
- (viii) The Contractor shall ensure that the drivers of all delivery vehicles to Site and those removing materials from Site wear minimum PPE of safety boots, safety helmet, safety eye wear, hi-visibility vest or uniform, safety gloves, and ear plugs at all times when they are outside their vehicle in a work area.
- (ix) The Contractor shall ensure that there is daily co-ordination of the movement of its mobile plant and vehicles on Site to minimize the potential for an incident.

- (x) All Site access roads used by mobile plant, equipment and vehicles shall be constructed of hard standing and suitable for its intended purpose.
 - (xi) No unauthorised parking shall be permitted on the Site anywhere other than at the designated parking area. Only cars bearing the S.O.'s permit shall be allowed to park there, and those of authorised visitors.
 - (xii) Site plant, equipment and vehicles shall be parked at designated locations only so that they do not cause obstructions to Site traffic.
 - (xiii) Consideration shall be given at all times to the safety of the road users and gates should be positioned so that they minimise the additional risk to traffic at such locations as road junctions, bends, etc.
 - (xiv) A clear line of sight shall be maintained for all drivers of vehicles using the road, taking into consideration the speed limit of that road and the position of existing street furniture and trees and vegetation, to maximise the warning distance of approach.
 - (xv) If deemed necessary by the S.O. suitable mirrors shall be positioned at these points to enhance vision of traffic movement both on the roadway and on entering / leaving the Site.
 - (xvi) Where the Site gates lead directly off a public street an effective method of controlling vehicles entering / leaving the Site shall be employed. This shall be manned by workers who have been trained in road safety and who are wearing specified PPE. Advice should be sought from the Traffic Police if necessary.
 - (xvii) If deemed necessary by the S.O. a traffic lane may be coned off to allow for safe deceleration.
 - (xviii) During the hours of darkness floodlighting of these Site entrances shall be provided to enhance visibility of such traffic controllers, but care must be taken that these lights do not dazzle any on-coming traffic or pedestrians or cause a nuisance to others. The traffic controllers should utilise hand held LED batons or gloves incorporating reflective material.
 - (xix) Warning signs in compliance with the Code of Practice for Traffic Control at Work Zone shall be conspicuously displayed at appropriate distances before such gates into the Site as to give all drivers a clear understanding of the traffic hazard ahead.
 - (xx) If vehicles entering or leaving the Site have to cross a public footpath or pavement then a worker must be deployed to control pedestrians as well as road traffic. Suitable warning signs should be deployed to alert pedestrians to possible traffic movement across the footpath.
- (b) Spillages on Roads
- (i) A paved truck wash bay for washing vehicles leaving the Site onto a roadway shall be provided and maintained at each vehicular egress point. Each truck wash bay design must be approved by the Board as part of the Earth Control Measures (ECM) Plan before it can be constructed on site.
 - (ii) Wash water from the wash bays shall be suitably treated before discharge or disposal.

- (iii) Contractor shall assign personnel and establish a system of checks to ensure that all vehicles and trucks leaving the worksite do not have the potential to litter the roads due to its wheels or transportation materials.
- (iv) Where it is foreseeable that water may drain out from a load of soft marine clay or similar transported wet materials that will contaminate the road surface, the Contractor shall ensure that specially designed and constructed watertight trucks are used to transport these materials.
- (v) All cement mixer trucks entering the site must have a containment system or a flap installed to prevent spillage of cement.

G2.23.56 Excavations and Deep Excavations

- (a) The Contractor shall appoint sufficient number of banksmen to coordinate excavation activities at the shafts/basements and the haulage activities from the shafts/basements to the at-grade level/bank. The banksman shall be appointed in writing and should have attended Signaller training course approved by the Ministry of Manpower (MOM). The banksman shall be properly identified on site, stationed at-grade and have overall control of the excavation works.
- (b) For excavations exceeding 4 m in depth, the Contractor shall appoint at least one banksman for every 30 m length of excavation. If deemed necessary by the S.O., the Contractor shall appoint additional banksmen. For excavations exceeding 10 m in depth, the Contractor shall appoint one banksman for every long-arm or telescopic excavator at the bank.
- (c) Communication in the form of walkie-talkie sets shall be established between the banksman and the excavator operators. The banksman shall be equipped with a loud hailer with siren alarm function to communicate during an emergency. No one shall be within any excavator's swing radius. All excavators shall be fitted with rear view camera for safe rear view monitoring.
- (d) Long arm excavators shall be provided with an extended reflective mirror in front of the operator's cabin to enhance operator's visibility and shall not operate without the presence of the banksman. All excavators shall also be fitted with a camera mounted on the long arm to further enhance the operator's view.
- (e) Excavator performing any excavation work in tunnels shall be equipped with spark arrester to prevent emission of flammable debris and naked light that will cause fire. A fire extinguisher with full capacity and in working order shall be placed in the operator's cabin.
- (f) Excavators within the excavation pit shall have suitably reinforced cabin roofs capable of withstanding impact from falling objects from the top of the excavation and its movement coordinated by one of its operator, who shall be appointed as a leader by the Contractor.
- (g) The designated locations at walers and struts, if any, used by the instrumentation contractor for instrumentation reading and monitoring shall be provided with two (2) rows or more of horizontal rigid guardrails to prevent persons falling from height. Openings within struts are to be covered. The vertical distance between the two (2) rows of horizontal guardrail shall be not more than 600mm. Toe boards are to be provided accordingly.

- (h) An alternative source of power and emergency lighting system shall be provided to allow emergency securing operations and evacuation safely in the event of a primary power failure. An adequate number of lamps shall be located at key points underground.
- (i) Where any worker in an excavation is exposed to the hazard of falling or sliding material from any bank or side more than 1.5 m high above his footing, adequate piling and bracing shall be provided against the bank or side to eliminate such hazard. The excavation shall be checked by a designated person after every rain, storm or other hazard increasing occurrence and the protection against slides or cave-ins shall be increased if necessary.
- (j) Excavated material and other superimposed loads shall be placed at least 1 m back from the edge of open excavations and trenches and shall be so shored or retained that no part thereof can fall into the excavation or cause the banks to slip or cause the upheaval of the excavation bed. Banks shall be stripped of loose rock or other materials which may slide, roll or fall upon persons below.

G2.23.57 Pipe Jacking

- (a) The Contractor shall ensure that no person enters a pipe of less than 1,200mm in diameter.
- (b) All work within a pipe jack shall follow strictly the procedures for work within a confined space and a Permit-to-Enter procedure shall be put in place and strictly followed.
- (c) All persons shall be out of the pipe when jacking is taking place and shall not re-enter until the ram is no longer in motion.

G2.23.58 Diaphragm Walls

- (a) The Contractor shall take the necessary measures to confirm if there are any services or utilities underground and potential soil movement of any unstable ground before constructing any guide walls for diaphragm wall activities.
- (b) The Contractor shall comply with the use of shoring to any excavation that exceeds 2 m. This may include the provision of hard barricade around the perimeter of the activity to avoid any fall.
- (c) The Contractor shall assess all excavation areas which are near to the vehicle carriageway or pedestrian footpath along the hoarding to minimize bentonite splatter. Checks on bentonite level and excavation depth shall be done regularly. Checks shall also be performed regularly on the bentonite quality including, but not limited to the viscosity, fluid loss and gel strength.
- (d) The Contractor shall ensure the presence of lifting supervisors, riggers signalmen and banksmen around the activity area before any chiselling, cutting and grabbing work commence. In any case of substantial ground movement, the Contractor is advised to move everyone and heavy machineries to a safe and stable ground.

- (e) The Contractor shall take extra measures during the desanding and cage installation activities, besides ensuring that all in-house safety procedures are followed thoroughly. This includes the assembling of tremie pipes where workers are to don at least a safety belt and be tied off to prevent any fall. Contractors shall ensure his lifting supervisors are competent in executing the task which not limited to checks on the 'U' bolts, the capacity of the crane and the crane lifting spreader bars.
- (f) Contractor shall maintain a site soil storage area to prevent from overflowing and assess the necessity in preventing of any spillage of slurry on the road or excessive spillage of soil during the transferring process.

G2.23.59 Overhead Hazards

- (a) The Contractor shall organise its work sequence to avoid to the extent practicable works underneath concurrent overhead works. Where this is not practicable, every place where persons are required to work or to pass that is normally exposed to falling material or objects shall be provided with suitable overhead protection. Where works are taking not taking place underneath but in the vicinity, the exposed area underneath the overhead works shall be barricaded or otherwise guarded to prevent inadvertent entry.

G2.23.60 Drowning Hazards

- (a) Where the work involves filling tanks with water leaving an open surface, the Contractor shall provide at all times and at suitable locations equipment for promptly rescuing persons from the water and resuscitating rescued persons.

G2.23.61 Slipping and Tripping Hazards

- (a) The Contractor shall keep all work areas and footpaths free of slipping and trip hazards. The Contractor shall not permit persons to use a passageway, scaffold, platform or other elevated working surface which is in slippery condition. Slip and trip hazards shall be promptly removed.

G2.23.62 Electrical Hazards

- (a) The Contractor shall ascertain by inquiry, direct observation or by instruments where any part of an electrical power circuit exposed or concealed is so located that the performance of the work may bring any person, tool or machine in physical or electrical contact therewith. The Contractor shall post and maintain proper warning signs of the location of such lines, the hazards involved and the protective measures to be taken and shall, if practicable, de-energize the electric power circuit.

G2.23.63 Eye Protection

- (a) Suitable eye protection equipment shall be provided for and shall be used by all workers while engaged in welding, cutting, grinding, chipping and any other works operation that may generate flying particles. Suitable eye protection shall also be provided and shall be used for all other works operation that may endanger the eyes.

G2.23.64 Environmental Management – General

- (a) The Contractor shall implement environmental management in accordance with General Specification G9.

G2.23.65 Hazardous Chemical Management

- (a) The Contractor shall establish and implement a programme to manage the receipt, movement, storage and use of hazardous chemicals in the site. The programme shall include:
 - (i) Written procedures for the acquisition, transportation, storage, usage and disposal of hazardous chemicals.
 - (ii) The appointment of a competent person to receive hazardous chemicals and manage their storage, distribution and handling.
 - (iii) Maintaining a register of hazardous chemicals together with their Material Safety Data Sheets (SDS). The SDS shall give details of the hazardous chemicals' compositions, physical and chemical properties, toxicity, flammability, personal protective equipment required during their handling, and measures to be taken during an emergency.
 - (iv) Labelling of all containers of hazardous chemicals. The labels shall indicate the hazards involved and the precautions to be taken.
 - (v) Designation of suitable storage areas. All unused hazardous chemicals shall be returned to the designated storage areas. Storage areas shall be secured against unauthorized access with storage and hazard notices clearly displayed.
 - (vi) Regular hygiene monitoring where there is exposure to airborne hazardous chemicals. Workers shall not be exposed to toxic airborne chemicals exceeding the permissible exposure level (PEL) specified in the applicable regulations.
 - (vii) Training and education of workers on the hazards associated with the hazardous chemicals, and the precautions to be taken during the storage, handling and use of these chemicals.
 - (viii) The provision, maintenance and use of PPE such as respirators. Workers shall be trained in the use of the PPE. The PPE shall be regularly inspected.
 - (ix) Arrangement of workers to undergo pre-employment and regular medical examinations if they are exposed to any of the chemicals listed in the applicable regulations such as WSH (Medical Examinations) Regulations.

G2.23.66 Post-Site Work Commencement Requirements

- (a) Safety Observations and Safety Inspections
 - (i) The Contractor shall provide and maintain a safe workplace and a safe system of work. The Contractor shall be responsible for conducting workplace inspections and safety observations. The Contractor shall make available copies of safety observation, safety inspections, accident statistics and near-miss report to the S.O. and the Board. Sample formats of reports are given in the Appendices.
- (b) Safety Dialogue
 - (i) The Contractor's Management and its WSH personnel shall attend periodic safety dialogue with the S.O. and the Board to review the Contractor's safety performance.

- (c) Site Supervisor
 - (i) The Contractor is required to provide the Board and the S.O. with the CV(s) before the employment of qualified competent full-time site supervisors according to the specifications. These supervisors must be safety-trained through attending the Building Construction Supervisor's Safety Course conducted by MOM's accredited service providers. The Contractor shall provide additional training as required for the Works.

G2.23.67 Safety Initiative and Feedback

- (a) The Contractor shall implement Safety Initiative and Feedback programmes for all his team including sub-consultants and sub-contractors. The key performance indicator for this programme shall be for one (1) suggestion for improvement on safety to be selected for implementation per 1000 man-hours worked. The Contractor shall include a summary report on Initiatives and Feedback implementation in the monthly safety report to the S.O..
- (b) The Contractor shall present a token of appreciation to the individual whose Safety Initiative and Feedback is successfully implemented on the project, as an encouragement for all workers to work safely.

G2.23.68 Appendices

- (a) APPENDIX G2.23A - Accident / Incident Occurrence Report Form.
- (b) APPENDIX G2.23B - Contractor's Monthly Safety, Health & Environment (HSE) Report.
- (c) APPENDIX G2.23C - Risk Assessment Guidelines.
- (d) APPENDIX G2.23D - Approved Codes of Practice.
- (e) APPENDIX G2.23E - Sample Report – Safety Observation Report by the Contractor.
- (f) APPENDIX G2.23F - Sample Report – Monthly Safety Inspection Report by the Contractor.
- (g) APPENDIX G2.23G - Sample Report – Monthly Accident Statistics Report by the Contractor.
- (h) APPENDIX G2.23H - Sample Report – Near-Miss Incident Report by the Contractor.
- (i) APPENDIX G2.23I - Photographs Highlighting Good HSE Practice.
- (j) APPENDIX G2.23J - Personal Protective Equipment.

END OF SECTION

APPENDIX G2.23A – Accident / Incident Occurrence Report Form

INCIDENT INVESTIGATION AND REPORT							
Incident Event & Involved/Injured Personnel Particulars							
Incident No.:		Name of Direct Employer:					
Incident Date/Time:		Employee Name:					
Date/Time Reported:		{Injured Person (IP)}					
Work Location: {for project work, please provide the project details and its contract sum}		Employee's Date of Birth / Age:					
		Employee's Nationality / ID No.:					
		Employee's Race:					
Work Elevation:		Employee's Skill Trade:					
Type of Incident Reported							
Non- Recordable Case:	☐ Near Miss	☐ Asset Damage	☐ Environmental	☐ First Aid			
Recordable Case:	☐ Dangerous Occurrence	☐ Medical Treatment Case	☐ Restricted Work Case				
Lost Time Incident:	☐ Occupational Disease	☐ Non- MOM Reportable Case	☐ MOM Reportable Case				
Involved Personnel Employment Details							
Length of Employment in Present Workplace	Year	Month	Length of Experience in Current Skill Trade	Year	Month		
Involved Witnesses / Implicated Persons / Accomplices Particulars							
Name of Direct Employer:			Name of Direct Employer:				
Concerned Person's Name:			Concerned Person's Name:				
Date of Birth / Age:			Date of Birth / Age:				
Nationality / ID No.:			Nationality / ID No.:				
Employee's Skill Trade:			Employee's Skill Trade:				
Year of Experience in Trade:			Year of Experience in Trade:				
Involved/Injured Employee's Direct Supervision & Occupier's Particulars							
Name of Direct Employer:			Occupier / Main Contractor:				
Direct Supervision Name:			Person-In-Charged Name:				
Supervision's DOB / Age:			PIC DOB / Age:				
Nationality / ID No.:			PIC Nationality / ID No.:				
Year of Experience in Trade:			PIC Designation:				
Supervision's Contact No.:			PIC Contact No.:				

First Aid & Administered Response Actions (if any)						
Name of OFA:		OFA ID No.:		Contact No.:		
☐ Emergency Wash	☐ Bandages	☐ Recovery Aid	☐ Administer CPR	☐ Administer AED		
☐ Hospital By Ambulance	☐ Hospital By Other Mode	☐ Sent to Hospital/Clinic:				
INCIDENT INVESTIGATION AND REPORT						
Chronological Sequence of Events Leading Till Incident Occurrence						
Work Shift Start Time:		Intended End of Work Shift:		Weather Condition:	☐ Fair ☐ Hazy ☐ Raining ☐ CAT1	
Timeline	Event Descriptions					
5-M Model Investigation Findings						
{Man: refers to human elements inclusive of physiological, psychological, proficiency aspects as well as qualifications of performing tasks}						
{Medium: refers to the environment where the task is to be conducted inclusive of weather conditions, terrain, obstructions, sunset/sunrise, lightning category, visibility, wind speed, etc.}						
<i>Cont'd...</i>						
{Machine: refers to the design, manufacturer instruction, maintenance of equipment/tools and engineering reliability / performance of equipment/structure/article}						
{Management: refers to supervisory capacity of management in terms of regulations, polices, attitudes towards safety}						
{Mission: refers to the type of task implemented, whether it was complex, routine, monotonous, repetitive, etc.}						

TUAS WATER RECLAMATION PLANT
CONTRACT 4B – DIGESTION PRE-TREATMENT AND
SIDESTREAM NITROGEN REMOVAL FACILITIES

VOLUME 2

Documentation Evidences Attachments		
☐ Risk Assessment/ Permit to Work	☐ Safe Work Procedures	☐ Work Method Statement
☐ MOM i-Report: *	☐ Job Safety Analysis	☐ Training Records
☐ IP/ Employee/ Witness Statement	☐ Safety Data Sheet	☐ Plans/Drawings/Pictures/Designs

* If MOM i-Reporting had been submitted, please fill in the notification number.

INCIDENT INVESTIGATION AND REPORT			
Nature of Injury Sustained (if any)			
☐ Bruises / Contusions	☐ Abrasions / Scratches	☐ Cut / Laceration	
☐ Burns (Heat/Chemical/Radiation)	☐ Puncture	☐ Fracture	
☐ Strain / Sprain / Dislocation	☐ Neurosis / Concussion / Comatose	☐ Amputation / Crushed	
☐ Dermatitis / Skin Rashes	☐ Toxicification / Poisoned	☐ Asphyxia / Drowning	
☐ Electrocution	☐ Blindness / Vision Impairment	☐ Fatal	
☐ Others or Multiple Injuries:			
Injured Body Parts (if any)			
☐ Skull / Scalp	☐ Face / Forehead	☐ Eyes	☐ Ears
☐ Nose	☐ Mouth / Throat / Teeth	☐ Neck	☐ Shoulder
☐ Chest	☐ Upper Back	☐ Lower Back	☐ Abdomen
☐ Pelvis / Hip	☐ Groin	☐ Upper Arm	☐ Forearm
☐ Elbow	☐ Wrist	☐ Hand / Palm / Finger	☐ Thigh
☐ Shin	☐ Knee	☐ Ankle	☐ Foot / Toes
☐ Internal Organ or Multiple Injuries:			
Medical Practitioner Diagnosis (if any or available)			
☐ Return to Work	☐ Light Duty	☐ Surgically Administered	☐ 3 Days or Less MC
☐ More than 3 Days MC	☐ Hospitalization for Less Than 24hrs	☐ Hospitalization for More Than 24hrs	
Incident Causations			
Direct Cause(s)			
Unsafe Condition(s) Found:			
Unsafe Act(s) Found:			
Work Process Discrepancy:			
Root Cause(s)			
Human Factors:			
Job Factors:			

Environmental Factors:		
Organizational Factors:		
INCIDENT INVESTIGATION AND REPORT		
Identified Gap(s) in OHS Management System		
Risk Management:		
Training / EHS Awareness:		
Communication:		
Legal Compliances:		
EHS Inspection / Audit:		
Maintenance Regime:		
Operational Control:		
Post Mortem from the Investigation Findings		
Emergency Response / Immediate Actions Taken Review		
Corrective Actions	Action By	Due Date
Preventive Measures	Action By	Due Date

TUAS WATER RECLAMATION PLANT
CONTRACT 4B – DIGESTION PRE-TREATMENT AND
SIDESTREAM NITROGEN REMOVAL FACILITIES

VOLUME 2

Post Investigation Follow-up Action Requirements		
☐ Review of Risk Assessment	☐ Review of Safe Work Procedure(s)	☐ Review of Training Manual
☐ Implement New Training	☐ Conduct Stand Down Briefing	☐ Implement New System of Work
☐ Initial Management Review	☐ Other Action(s):	
Prepared By:	Reviewed By:	Acknowledged By:
Date & Signature	Date & Signature	Date & Signature
Name & Designation:	Name & Designation:	Name & Designation:

APPENDIX G2.23B – Contractor's Monthly Safety, Health & Environment (HSE) Report

The report shall follow the format given:

1. Project Profile

A brief description of the project e.g. Contract Title, Contract No., Award Date, Completion Date etc.

2. Executive Summary

To give a brief summary of the site HSE events such as any educational, promotional and enforcement activities.

APPENDICES

The following items shall be attached. Nil return to be indicated if the section is not applicable:

- a) Contractor's Monthly Incident / Accident Return Summary.
- b) Contractor's Monthly Accident Statistics Report Form.
- c) Incident Listing - A listing of all incidents/accidents since the beginning of the project with analysis - trend of incidents, root causes, preventive measures etc.
- d) Updated organisation chart.
- e) Title listing of all existing method statements.
- f) Training register based on locations within the contract indicating completed and scheduled.
- g) Summary of all toolbox meetings.
- h) Listing of HSE promotional activities carried out during the month.
- i) Schedule of HSE promotional activities for next 6 months.
- j) Summary of evaluation carried out on the HSE performance of sub-contractors with actions taken.
- k) Copies of HSE reports to MOM / NEA, HSE related meetings & WSHO / ECO inspection reports with photos, close-out actions and target date of completion.
- l) Register of plants, equipment and tools requiring statutory inspection with expiry dates and date of next inspection register of all plants, equipment and tools with monthly maintenance dates and date of next maintenance.
- m) Register of hazardous substance & chemicals with their hazards, location and inventory.

- n) Listing of emergency drills since the beginning of the project.
- o) Summary of Gas monitoring for the month with reasons for exceeding the limits.
- p) Audit Schedule and update of the last external safety audit action plan.
- q) Listing of HSE Awards since project commencement.
- r) HSE initiatives and best practices (to attach photos)
- s) Summary of visits by MOM, NEA, PUB or other authorities and its outcome such as fines, warning notices and demerit points. Brief description of follow up actions carried out pertaining to non-conformances found during the visits. -
- t) Listing of environmental non-compliance received since project commencement, including short description of non-compliance, date of occurrence, date of fine received, fine amount and corrective and/or preventive action(s) taken, etc.
- u) Title listing of all environmental management plans.
- v) Implementation status of environmental controls, including the following information for the environmental aspects listed from part A to F below:
 - Upcoming activities that will have impact on stakeholders on the following aspects;
 - Current and upcoming environmental protection and mitigation measures on the following aspects (include site layout, plans, implementation status where applicable);
 - Inspection and maintenance regime/schedule.

Environmental Aspects

- A. Earth Control.
- B. Noise Pollution.
- C. Vector Control.
- D. Air Pollution.
- E. Construction Waste Management.
- F. Water Management and Conservation.

- w) Summary of public feedback/complaints, and the investigation and action taken to address the issue.

- x) Summary of noise monitoring data for the month (in weekly graphical comparison format against the permissible noise limits) with reasons for exceeding the limits.
- y) Summary of TSS monitoring results for the month with reasons for exceeding the limits.
- z) Monthly ECM Inspection Report conducted by QECP.

ECO Weekly & THE S.O. / S.O.R. Fortnightly Environmental Inspection Reports.

APPENDIX G2.23C – Risk Assessment Guidelines

1. The risk assessment process normally requires baseline information on appropriate control measures, derived from a number to which the Contractor is expected to make reference:
 - Industry standards and codes of practice.
 - Experience of individuals.
 - Accidents/incidents records.
 - Knowledge of processes/activities.
2. It is essential that these risk assessments are carried out by teams of appropriately experienced and competent people.

3. General Risk Assessment Procedure

3.1 Breaking Down the Activity into its Constituent Elements

To breakdown an activity the following considerations must be taken into account:

- A step by step breakdown of the work activity.
- The persons involved in the work.
- The duration of work.
- The type of equipment/machinery.
- The type of material handled.
- The work environment.

3.2 Identifying the Hazards associated with each step

Consider what could go wrong at each step-in order to identify the types of hazards in the work activity and Critical Risks with controls to be separately listed:

- The hazards a person is exposed to at every step of the work activity, e.g. Falling from height, material falling from height, slipping.
- The hazards associated in operating the equipment / machinery.
- The hazards associated in handling the material.
- The hazards relating to the work environment.

3.3 Evaluate Accident Frequency

3.3.1 This is based on the judgement of those carrying out the assessment and where possible supported by The Contractor's past projects statistics.

3.3.2 **TABLE 1:** Accident Frequency

Level	Likelihood	Description	Probability
1	Rare	Not expected to occur but still possible	Unlikely, but may exceptionally occur
2	Remote	Not likely to occur under normal circumstances	Likely to occur once in 5-year project period
3	Occasional	Possible or known to occur	Likely to occur once a year
4	Frequent	Common occurrence	Likely to occur 4 times per year
5	Almost Certain	Continual or repeating experience	Likely to occur 12 times or more per year

3.4 Evaluate Accident Severity

3.4.1 An estimation of the likely consequences of the hazardous element occurring in terms of injury or loss.

3.4.2 **TABLE 2:** Accident Severity

Rating	Consequence	Description (*)
5	Catastrophic	- Single or Multiple loss of life from injury or occupational disease, immediately or delayed; and/or - Loss of whole production for greater than 3 days and/or - Total loss in excess of \$1 million.
4	Critical/Major	- Reportable major injury, occupational disease or dangerous occurrence; and/or - Damaged to works or plants causing delays of up to 3 days; and/or - Total loss in excess of \$100,000 but up to \$1 million.
3	Marginal/Moderate	- Reportable injury, occupational disease and/or - Damage to works or plants causing delays of up to 1 man-day; and/or, - Total loss in excess of \$10,000 but up to \$100,000.
2	Minor	- Minor injury, no lost time or person involved returns to work during the shift after treatment; and/or - Damage to works or plants does not cause significant delays; and/or - Total loss of between \$ 1,000 to \$10,000.
1	Negligible	- Not likely to cause injury or property damage - Low financial loss below \$1,000

3.5 Evaluate the Risk Matrix

3.5.1 TABLE 3: Risk Index Matrix

Likelihood Severity	Rare (1)	Remote (2)	Occasional (3)	Frequent (4)	Almost Certain (5)
Catastrophic (5)	5	10	15	20	25
Major (4)	4	8	12	16	20
Moderate (3)	3	6	9	12	15
Minor (2)	2	4	6	8	10
Negligible (1)	1	2	3	4	5

3.5.2 The definitions of the risk indices determined from the Risk Index Matrix are presented in the table 4.

3.5.3 TABLE 4: Definition of Risk Index

Risk Level	Risk Acceptability	Recommended Actions
Low Risk (1-3)	Acceptable	- No additional risk control measures may be needed. - Frequent review and monitoring of hazards are required to ensure that the risk level assigned is accurate and does not increase over time.
Medium Risk (4-12)	Tolerable	- A careful evaluation of hazards should be carried out to ensure that the risk level is reduced to as low as reasonably practicable within a defined period. - Interim risk control measures, such as administrative controls or PPE, may be implemented while longer term measures are being established. - Management attention is required.
High risk (15-25)	Not Acceptable	- High-risk level must be reduced to at least medium risk before work commences. - There should not be any interim risk control measures. Risk control measures should not be overly dependent on PPE or appliances. - If practicable, the hazard should be eliminated before work commences. - Management review is required before work commences.

Appendix G2.23D – Approved Local Codes of Practice referred to:

	<i>Title</i>	<i>Year Published</i>
1.	Code of Practice for Working Safely at Heights	2013
2.	Code of Practice on WSH Risk Management	2015
3.	Code of Practice on Safe Lifting Operations in the Workplaces	2014
4.	SS 98: Specification for industrial safety helmets	2013
5.	SS 280: Specification for Metal scaffoldings	2006
	Part 1: Frame scaffoldings	
6.	SS 280: Specification for Metal scaffoldings	2009
	Part 2: Modular scaffoldings	
7.	SS 311: Specification for Steel tubes and fittings used in tubular scaffolding	2005
8.	SS 343: Specification for lifting gear — Part 1: Wire rope slings	2014
9.	SS 343: Specification for lifting gear — Part 2: Hooks	2014
10.	SS 343: Specification for lifting gear — Part 3: Shackles	2014
11.	SS 473: Specification for Personal eye-protectors	2011
	Part 1: General requirements	
12.	SS 473: Specification for Personal eye-protectors	2011
	Part 2: Selection, use and maintenance	
13.	SS 497: Code of Practice for Design, safe use and maintenance of gantry cranes, overhead travelling cranes and monorail hoists	2011
14.	SS 506: Occupational safety and health (OSH) management system	2009
	Part 1: Requirements	
15.	SS 506: Occupational safety and health (OSH) management system	2009
	Part 2: Guidelines for the implementation of SS 506: Part 1: 2009	
16.	SS 506: Occupational safety and health (OSH) management system	2013
	Part 3: Requirements for the chemical industry	
17.	SS 508: Graphical symbols - Safety colours and safety signs	2013
	Part 1: Design principles for safety signs and safety markings	
18.	SS 508: Specification for graphical symbols — Safety colours and safety signs	2008
	Part 2: Design principles for product safety labels	
	Amendment No.1 (2013)	

	<i>Title</i>	<i>Year Published</i>
19.	SS 508: Graphical symbols — Safety colours and safety signs	2013
	Part 3: Design principles for graphical symbols for use in safety	
	signs	
20.	SS 508: Graphical symbols — Safety colours and safety signs	2013
	Part 4: Colorimetric and photometric properties of safety sign	
	materials	
21.	SS 508: Graphical symbols — Safety colours and safety signs	2013
	Part 5: Registered safety signs	
22.	SS 510: Code of Practice for Safety in welding and cutting (and	2017
	other operations involving the use of heat)	
23.	SS 511: Code of Practice for Diving at work	2010
24.	SS 513: Specification for personal protective equipment —	2013
	Footwear	
	Part 1: Safety footwear	
25.	SS 513: Specification for personal protective equipment —	2013
	Footwear	
	Part 2: Test methods for footwear	
26.	SS 514: Code of Practice for Office ergonomics	2016
27.	SS 528: Specification for Personal fall-arrest systems	2006
	Part 1: Full-body harnesses	
28.	SS 528: Specification for Personal fall-arrest systems	2006
	Part 2: Lanyards and energy absorbers	
29.	SS 528: Specification for Personal fall-arrest systems	2006
	Part 3: Self-retracting lifelines	
30.	SS 528: Specification for Personal fall-arrest systems	2006
	Part 4: Vertical rails and vertical lifelines incorporating a	
	sliding-type fall arrester	
31.	SS 528: Specification for Personal fall-arrest systems	2006
	Part 5: Connectors with self-closing and self-locking gates	
32.	SS 528: Specification for Personal fall-arrest systems	2006
	Part 6: System performance tests	
33.	SS 531: Code of Practice for Lighting of work places	2006
	Part 1: Indoor	
34.	SS 531: Code of Practice for Lighting of work places	2008
	Part 2: Outdoor	

	<i>Title</i>	<i>Year Published</i>
35.	SS 531: Code of Practice for Lighting of work places	2008
	Part 3: Lighting requirements for safety and security of outdoor work places	
36.	SS 532: Code of Practice for the storage of flammable liquids	2016
	Incorporating Corrigendum No. 1	
37.	SS 536: Code of Practice for the safe use of mobile cranes	2008
	(Formerly CP 37)	
38.	SS 537: Code of Practice for Safe use of machinery	2008
	Part 1: General requirements	
39.	SS 537: Code of Practice for Safe use of machinery	2009
	Part 2: Woodworking machinery	
40.	SS 541: Restraint belts	2008
	Amendment No.1 (2012)	
41.	SS 548: Code of Practice for Selection, use, and maintenance of	2009
	respiratory protective devices	
	(Formerly CP 74)	
42.	SS 549: Code of Practice for Selection, use, care and	2009
	maintenance of hearing protectors	
	(Formerly CP 76)	
43.	SS 550: Code of Practice for Installation, operation and	2009
	maintenance of electric passenger and goods lifts	
	(Formerly CP 2)	
44.	SS 553: Code of Practice for air-conditioning and mechanical	2016
	ventilation in buildings	
45.	SS 554: Code of Practice for indoor air quality for	2016
	air-conditioned buildings	
46.	SS 557: Code of Practice for Demolition	2010
	(Formerly CP 11)	
47.	SS 559: Code of Practice for Safe use of tower cranes	2010
	(Formerly CP 62)	
48.	SS 562: Code of Practice for Safety in trenches, pits and other	2010
	excavated areas	
49.	SS 567: Code of Practice for Factory layout — Safety, health	2011
	and welfare considerations	
	(Formerly CP 27)	
50.	SS 568: Code of Practice for Confined spaces	2011
	(Formerly CP 84)	

	<i>Title</i>	<i>Year Published</i>
51.	SS 569: Code of Practice for Manual handling (Formerly CP 92)	2011
52.	SS 570: Specification for Personal protective equipment for protection against falls from a height — Single point anchor devices and flexible horizontal lifeline systems	2011
53.	SS 571: Code of Practice for Energy lockout and tagout (Formerly CP 91)	2011
54.	SS 573: Code of Practice for The safe use of powered counterbalanced forklifts (Formerly CP 101)	2012
55.	SS 580: Code of Practice for Formwork (Formerly CP 23)	2012
56.	SS 586: Specification for hazard communication for hazardous chemicals and dangerous goods Part 1: Transport and storage of dangerous goods	2014
57.	SS 586: Specification for hazard communication for hazardous chemicals and dangerous goods Part 2: Globally harmonised system of classification and labelling of chemicals — Singapore's adaptations	2014
58.	SS 586: Specification for hazard communication for hazardous chemicals and dangerous goods Part 3: Preparation of safety data sheets (SDS) Amendment No. 1 (2014)	2014
59.	SS 588: Personal equipment for protection against falls — Rope access systems Part 1: Fundamental principles for a system of work	2013
60.	SS 588: Personal equipment for protection against falls — Rope access systems Part 2: Code of practice	2013
61.	SS 595: Steel wire ropes for hoisting — Part 3: Code of practice for the care, inspection and maintenance of steel wire ropes for hoisting	2014
62.	SS 598: Code of Practice for Suspended scaffolds	2014
63.	SS 607: Specification for design of active fall-protection systems	2015
64.	CP 5: Code of Practice for Electrical installations Amendment No. 1 (2008)	1998
65.	CP 14: Code of Practice for Scaffolds	1996
66.	SS 617: Code of Practice for the lifting of persons in work platforms suspended from cranes	2016

	<i>Title</i>	<i>Year Published</i>
67.	CP 79: Code of Practice for Safety management system for construction worksites	1999
68.	CP 88: Code of Practice for Temporary electrical installations Part 1: Construction and building sites	2001
69.	CP 99: Code of Practice for Industrial noise control	2003

International HSE Laws and Codes of Practice referred to:

1. Australian and New Zealand Standards.
2. European Harmonised Standards.
3. OSHA Standards.
4. British Standards.

Appendix G2.23E - Sample Report

Safety Observation Report by Contractor

Date and Time: _____ Report No.: _____
 Worksite Location: _____
 Reference No: _____ Project Officer: _____
 Contract No: _____
 Contract Title: _____
 Contractor Name: _____
 Contract Period: From _____ to _____

Criteria (more may be added as necessary)

Site Organisation			
Observations indicate that:	Yes	No	NA
• Work areas are properly barricaded?			
• Site/area is free from rubbish and obstructions?			
• Openings in floors, trenches etc are covered or barricaded?			
Tools and Equipment			
Observations indicate that:	Yes	No	NA
• Tools and equipment are right for the job?			
• Tools and equipment are used correctly?			
• Tools and equipment are in good condition?			
• Insulated ladders are used near live exposed electrical equipment?			
• Portable residual current devices are used, where required?			
Prevention of Falls from Height			
Observations indicate that:	Yes	No	NA
• Elevated work platforms have handrails & kickboards?			
• Harnesses with lanyards being used?			
• Evidence of damage to ladders?			
Personal Protective Equipment			
Observations indicate that:	Yes	No	NA
• Relevant Personal Protective Equipment is used?			

Remarks _____

Observations by: _____
 Name / Designation (WSH Personnel) Signature
 Approved by Project Manager: _____
 Name / Designation / Date Signature

Note: At least one safety observation of the complete Site is to be conducted each week. A copy of this safety observation report is to be submitted to the S.O. by the following week

Appendix G2.23F - Sample Report

Monthly Safety Inspection Report by the Contractor

Month / Year: _____ Report No: _____
Contract No: _____
Contract Title: _____
Contractor Name: _____
Contract Period: From _____ to _____

SECTION A - Inspection Details

Reference No: _____ Project Officer: _____
Date and Time: _____
Worksite Location: _____
Inspection Team Leader: _____
Inspection Team Members (to include WSH personnel): _____

SECTION A - Findings & Recommendations

S/N	Subject	Findings & Recommendations	Action by

Remarks _____

Submitted by Team Leader _____
Name / Designation / Date _____ Signature _____

Accepted by Contractor's Project Manager _____
Name / Designation / Date _____ Signature _____

Note: A copy of this report is to be submitted to the S.O. by the first week of the following month

Appendix G2.23G - Sample Report

Monthly Accident Statistics Report by the Contractor

Month / Year: _____ Report No: _____
 Contract No: _____
 Contract Title: _____
 Contractor Company Name: _____
 Contract Period: From _____ to _____

SECTION A - Accident Statistics

	Target	Last Month	This Month	Cumulative*
No. of workers				
No. of man-hours worked				
No. of lost days				
No. of lost-time injury				
Accident Frequency Rate (AFR)				
Accident Severity Rate (ASR)				
No. of Occupational Disease Incidents (ODI)				
Occupational Disease Incident Rate				

* From the first month when site work commences.

Accident Frequency Rate = $\frac{\text{No. of lost-time injury} \times 10^6}{\text{No. of man-hours worked}}$

Accident Severity Rate = $\frac{\text{No. of lost-days} \times 10^6}{\text{No. of man-hours worked}}$

Occupational Disease Incidence Rate = $\frac{\text{No. of ODI cases} \times 100,000}{\text{No. of persons employed}}$

<i>Serious violations reported by the Contractor to MOM. Stop Work Order and Fine are those imposed by any relevant Authorities relating to or in connection with work, safety or health of the employees or sub-contractors. Details are to be attached to report.</i>			
Types		No. of Cases	
		Last Month	This Month
Dangerous Occurrence (DO) as specified in the 2 nd Schedule of WSH Act.			
Property Damage incident			
Stop Work Order			
Fine			
Remarks			
Submitted by WSH Coordinator/Officer	_____		_____
	Name / Designation / Date		Signature
Approved by Contractor 's Project Manager	_____		_____
	Name / Designation / Date		Signature
<i>Note: A copy of this report is to be submitted to the S.O. by the first week of the following month</i>			

Appendix G2.23H - Sample Report

Near-Miss Incident Report by the Contractor

Report No: _____

Contract No: _____

Contract Title: _____

Contractor Name: _____

Contract Period: From _____ to _____

Reference No: _____ Project Officer: _____

SECTION A - Report

To Project Manager

I witnessed a near-miss incident on _____ at _____
(date) (time)

at _____
(location / installation)

Brief description of incident
(what I saw and happened)

Possible injuries or damages that could have arisen
(to person, object, material and environment)

Recommendation

Reported by:

Name* Date: _____

* Optional

SECTION B - Follow-up

Follow-up by:

Name / Designation / Date Signature

Action Taken:

Note: Near-miss incident is an event which under slightly different circumstances, could result in injury to person(s) or damage to plant, material or environment.

Appendix G2.23I – Photographs Highlighting Good HSE Practice to be adopted

The following photographs are included for reference purposes to assist the Contractor in developing its safety and health systems and procedures. They complement the requirements in applicable statutes, regulations, standards or other requirements defined in these specifications.



Examples of Step Platforms



Examples of Step Platforms being used on site



Steel staircase with unobstructed headroom and netting wrapped around handrail and underneath of steps to prevent loose material from falling through



Handrails (top and intermediate)

Proper handrails provided at designated site access and walkways. Toe board and netting are added to prevent loose material from falling over



Handrails (top and

Proper handrails 1.3 m height provided at designated site access and walkways. Toe board are added to prevent loose material from falling over. Netting added to prevent loose material from falling



Mobile machineries such as Cranes, Excavators and trailer beds to be fitted with fixed or removable Guard rail-Hand rails and/or Fall mitigation controls to prevent personals from falling-slipping from height > 1.8m



Floor opening cover is flushed and secured against moving



Lifting activity is cordoned off and warning signage is placed to warn personnel against entering the area





Lifting gears properly stored and maintained in good condition

MONTH	COLOUR
January, May & September	GREEN
February, June & October	WHITE
March, July & November	BLUE
April, August & December	YELLOW
Defective items / Due Servicing	RED

Colour code system implemented to ensure lifting gear is check thoroughly every month before use



Ventilation fan with fixed mechanical guard to prevent fingers from coming into contact the moving fan blades



Sticker label to indicate that powered tool has been checked monthly by LEW



Powered drill with Active Torque Control (ATC) safety feature (or equivalent) to prevent uncontrolled spinning of drill when drill bit gets jammed. The uncontrolled spinning and drill bits breaking can cause injury to operator



cable hangers

Electrical cable hangers insulated to be used



Flammable substance stored at designated storage area with warning signage. Area is locked to prevent unauthorized removal of substance



Gas cylinders secured in a cylinder trolley to prevent toppling



Firefighting and first aid equipment readily available at strategic locations



Height Limit Gantry installed at site entrance/exit to ensure the total height of the vehicle does not exceed 4.5m



Site entrance equipped with revolving light and convex mirror to enhance pedestrian safety. Full-time banksman deployed to guide vehicles exiting the worksite



Overhead shelter provided when work is carried out near hoarding to protect persons from any falling objects



“See through fencing” allows pedestrian to check and ensure vehicles has stopped before crossing at traffic junction



Good housekeeping at worksite with designated areas for storage and walkway



Good housekeeping at worksite with designated areas for storage and walkway



**Delineating area
for storage and walkway**

**Good housekeeping at worksite with designated areas
for storage and walkway**



**Good housekeeping. Material storage area separate
from walkway**



Proper First Aid room equipped with standard first aid items, stretcher and bed



Proper rest areas with tables and chairs provided



Proper rest areas with tables and chairs provided



Adequate Water cooler provided at worksite. Water Filter checked and maintained monthly.



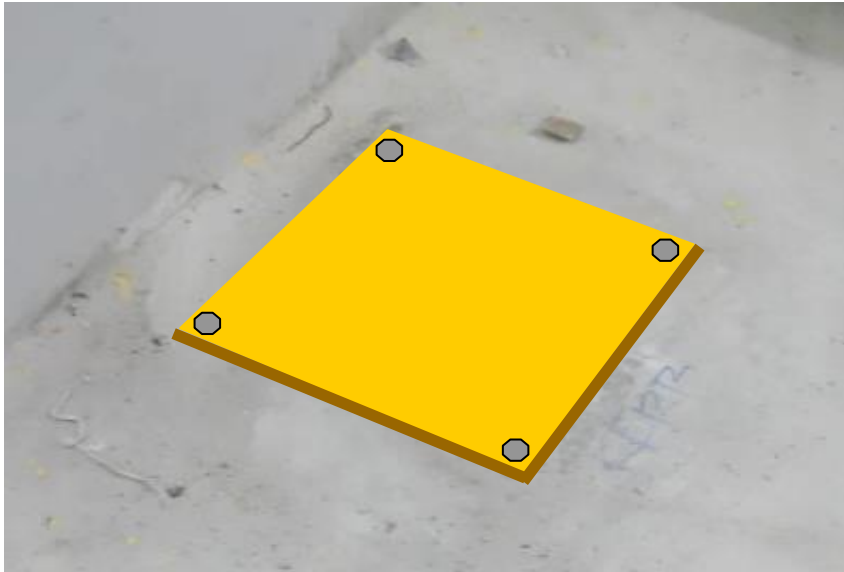
Stage provided for safety promotional events



Safety promotion corner – Display of exhibits to communicate safety requirements



Coloured Safety posters/articles displayed to communicate safety requirements



Note: Yellow was chosen because is distinctive, but any colour or pattern can be adopted.



A13 Mesh Cast into Slab

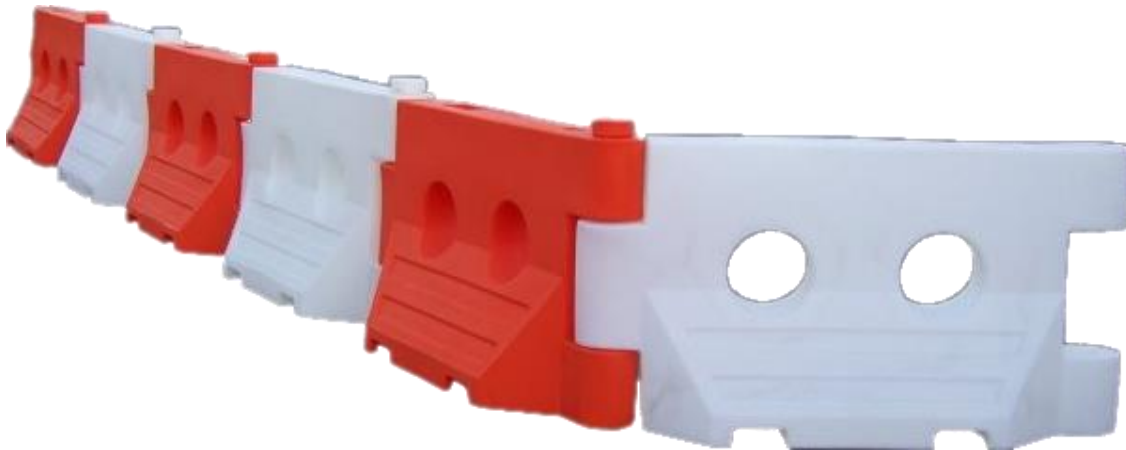
Mesh cast with slab provides immediate protection against fall and cannot be accidentally removed.



Barricade shown above is meant to protect against people. It is not appropriate for use where vehicle movement is expected. A properly designed steel structure is recommended in such situations.



Sliding gate restraint from movement by securing to gate post.



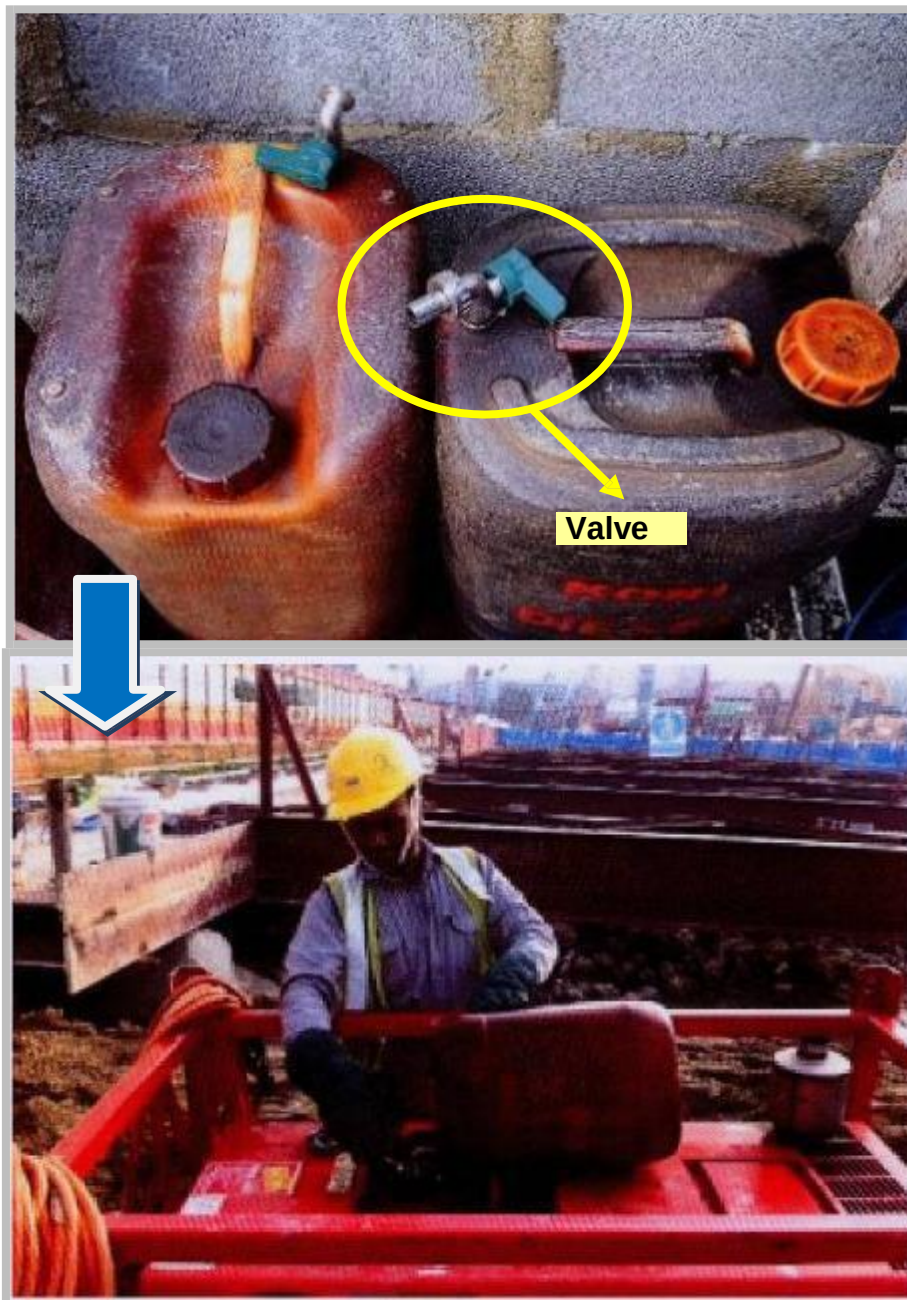
The right type of barriers to be used to set up cordoned work area and Traffic



Top opening of treatment plant (that is temporarily not in use) is covered with anti-mosquito netting



Provision of adequate and sheltered bins for containment of wastes



Use of valve during diesel refill

By having this valve, diesel refill can be done in a more controllable manner, thus. preventing spillage.



Use of 'green' hoardings to beautify the site and to reduce the impact from traffic carbon emissions

Appendix G2.23J - General PPE Matrix for TWRP DTSS 2

The Contractor shall determine and provide the correct types of PPE to suit specific tasks.
The images are for illustration and identification purposes.

PPE Type	Specification(s)	All Contractors	Visitor
Hard Hat with Ratchet/Chin strap	Spec – SS 98		
Safety Glasses	Spec - SS 473, ANSI Z87.1		Any Approved Type as per specification
Safety Glasses (Over Spec)	Spec - SS 473, ANSI Z87.1		Any Approved Type as per specification
Ankle High Lace up / Zip up / Elastic Steel Cap Safety Boots	SS 513 AS/NZS 2210.3		Any Approved Type as per specification
6" High Zip up Steel Cap Safety Boots	SS 513 AS/NZS 2210.3		Any Approved Type as per specification

PPE Type	Specification(s)	All Contractors	Visitor
Hand Gloves Cut Resist grade (CE Level 5) Must be Appropriate for the task	Spec – SS EN 420, EN 388 CE cut level 5		Any Approved Type as per specification
Hearing Protection of 25 Decibels NRR Must be Appropriate for the task	3M with box Spec – SS 549 Or Disposable Foam ear plugs		Any Approved Type as per specification
Fall Protection Gear Harness with Lanyard (Fixed or Retractable) not exceeding 1 m in length	Spec – SS 528 Parts 1- 6, ANSI Z359		Visitor not allowed to work at height
Fall Protection – Suspension trauma strap (1 pair)	Spec – SS 528 Parts 1- 6, ANSI Z359		Visitor not allowed to work at height

PPE Type	Specification(s)	All Contractors	Visitor
Long Sleeve & Long Pants High Visibility (Class 2) Contractor shall determine their color of uniform including Sub-contractors. 100% Cotton 40-50+ SPF coated Rear & under arm Ventilated Fabric GSM below 150 Visitor vest are excluded from the specification.	Australian (AS) / New Zealand Standards (NZS): • AS/NZS 2919 • AS/NZS 4399 • AS/NZS 1020 • AS/NZS 4602 • AS/NZS 1906 European (EN) Standards: • EN 340 • EN 1149-3 • EN 13034 • EN 471 3M Reflective Taping		 

PPE Type	Specification(s)	All Contractors	Visitor
Long Sleeve & Long Pants High Visibility (Class 2) Contractor shall determine their color of uniform including Sub-contractors. 40-50+ SPF coated FR label to be externally visible    	Flame and Heat resist garment specification • NFPA 2112 • NFPA 70E HRC CAT 2 • EN ISO 11612 • EN ISO 14116 • ASTM F1506 • ASTM D6413 Australian (AS) / New Zealand Standards (NZS): • AS/NZS 2919 • AS/NZS 4399 • AS/NZS 1020 • AS/NZS 4602 • AS/NZS 1906 European (EN) Standards: • EN 340 • EN 1149-3 • EN 13034 • EN 471 • 3M flame resist Reflective Taping		Visitor not allowed to Hazardous areas

Helmet Color	Vest or Uniform Color	Appointment
WHITE: 	Refer to Uniform requirement under PPE matrix	For Engineers, Managers, Supervisors and Foreman
BLUE: 	Refer to Uniform requirement under PPE matrix	For Health, Safety and Environmental Personals
RED: 		Rigger - Signalman

Helmet Color	Vest or Uniform Color	Appointment
RED: 		Banksman – Spotter – Traffic Marshal
BROWN: 		Lifting Supervisor
GREEN: 		Welder

Helmet Color	Vest or Uniform Color	Appointment
YELLOW: 	Refer to Uniform requirement under PPE matrix	For General Labourers and Machinery Operators such as Excavator, Crane, Earth Moving, Rollers and equivalent